

相对论性能量和动量：高速运动

(1) **Total energy 总能量** $E = mc^2$

(2) **Rest energy 静止能量** $E = m_0 c^2$

(3) **K : relativistic kinetic energy 相对论动能**

$$K = mc^2 - m_0 c^2$$

(4) **Relation between the total energy E and momentum p**

$$p^2 c^2 = E^2 - E_0^2$$

$$p = \frac{1}{c} \sqrt{E^2 - E_0^2} = \frac{1}{c} \sqrt{(E + E_0)(E - E_0)} = \frac{1}{c} \sqrt{K(K + 2E_0)}$$

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Key words And Terms

black body radiation

黑体辐射

the photoelectric effect

光电效应

ultraviolet light

紫外光

threshold (cutoff) frequency

截止频率

stopping voltage or stopping potential

遏止电压

work function

功函数

photon

光子

Compton scattering

康普顿散射

Compton wavelength

康普顿波长

discrete

不连续的

duality

二象性

Chapter 22 Early Quantum Theory and Model of the Atom

22.1 Black Body Radiation and Planck's Quantum Hypothesis

22.2 The Photoelectric Effect

22.3 Compton scattering

22.4 The Bohr model of hydrogen atom

22.5 Matter waves

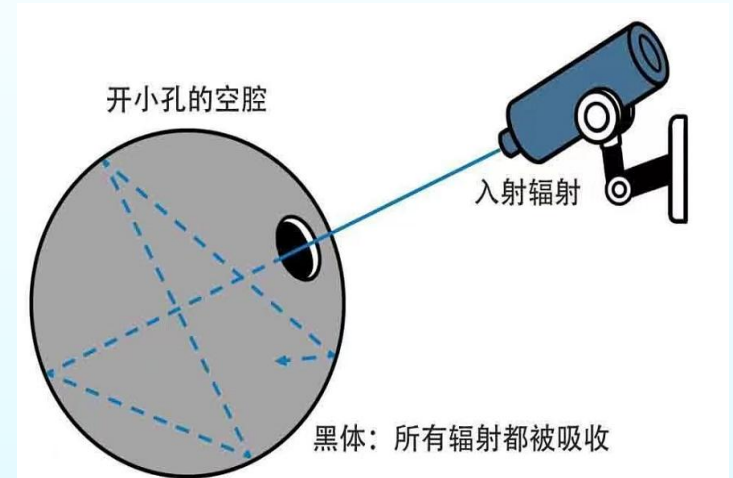
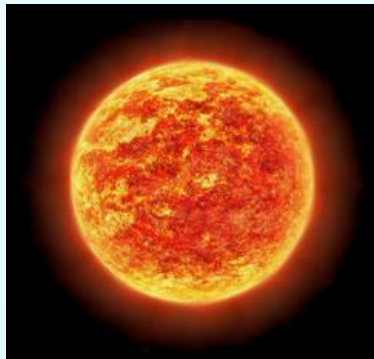
22.1 Black Body Radiation and Planck's Quantum Hypothesis

1. Black body radiation

Heat radiation: **Every** body emits **all** kinds of electromagnetic waves at **any** time.任何物体在任何温度下都在不断地向外发射各种波长的电磁波.

Blackbody (idealized blackbody): A body would absorb all the radiation falling on it.

黑体模型：设有一空心容器，器壁由不透明材料制成，器壁上开有一小孔O，小孔可看做黑体。



Blackbody radiation

Among objects listed below, () belongs to **absolute blackbody**



(A) object not emitting visible light

(B) object not emitting any light

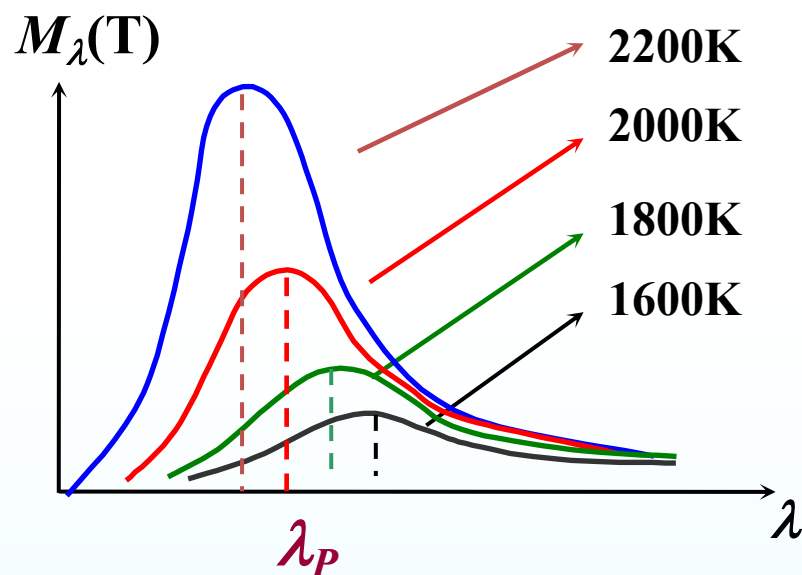
(C) object cannot reflect visible light

☒ (D) object cannot **reflect** any light

2. Law of black body radiation

黑体的单色辐出度 $M_\lambda(T)$ 在温度一定时随波长 λ 的变化实验规律。

$$M(T) = \int_0^\infty M_\lambda(T) d\lambda$$



(1) Stefan-Boltzmann's law 斯特藩-玻耳兹曼定律

$$M(T) = \sigma T^4 \quad \sigma = 5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^{-4}$$

(2) Wien's law 维恩位移定律

$$T \lambda_p = b \quad b = 2.90 \times 10^{-3} \text{ m} \cdot \text{K}$$

3 . Planck's Quantum Hypothesis

(1) Planck's theory

a. Wien's theory (1893)

$$M_{\lambda}(T) = C_1 \lambda^{-5} e^{-\frac{C_2}{\lambda T}}$$

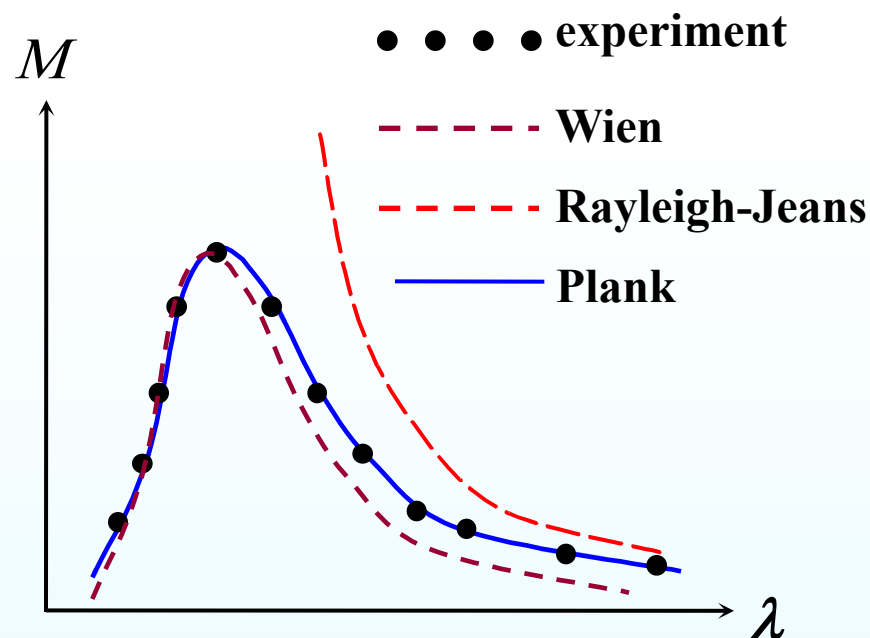
b. Rayleigh-Jeans theory (1900-1905)

$$M_{\lambda}(T) = C_2 \lambda^{-4} T$$

c. Planck's theory (1900)

$$M_{\lambda}(T) = \frac{2\pi hc^2 \lambda^{-5}}{e^{\frac{hc}{\lambda T}} - 1}$$

Planck's constant: $h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$



Planck's theory

$$M_{\lambda}(T) = \frac{2\pi hc^2 \lambda^{-5}}{e^{hc/kT\lambda} - 1}$$

(1) λ is large

$$e^{hc/kT\lambda} - 1 \rightarrow hc / kT\lambda$$

$$M_{\lambda}(T) = 2\pi^2 \lambda^{-5} \times hc / kT\lambda = 2\pi^2 \lambda^{-4} T$$

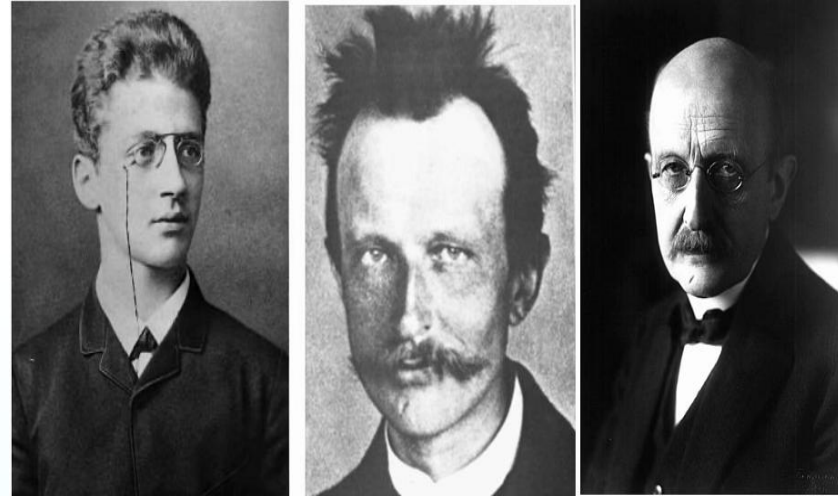
(2) λ is small

$$e^{hc/kT\lambda} - 1 \rightarrow e^{hc/kT\lambda}$$

$$M_{\lambda}(T) = \frac{2\pi hc^2 \lambda^{-5}}{e^{hc/kT\lambda} - 1} = \frac{2\pi hc^2 \lambda^{-5}}{e^{hc/kT\lambda}} = 2\pi hc^2 \lambda^{-5} e^{-hc/kT\lambda} = C_1 \lambda^{-5} e^{-\frac{C_2}{\lambda T}}$$

(2) Planck's Quantum Hypothesis

The energy distributed among the oscillations of atoms within molecules is **not continuous**, but instead consists of a finite number of very small **discrete** amounts. The **minimum energy** is



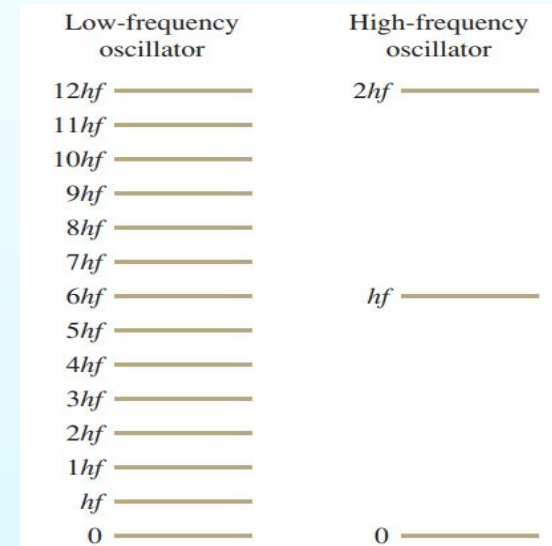
M. Planck (1858--1947)

1918 "Nobel Prize"

$$E_{\min} = hf \quad \text{quantum of energy (能量量子)}$$

Other energy of any molecular vibration could be only some **integer** times hf .

$$E = nhf \quad n=1,2,3,\dots \quad \text{quantized}$$



Example (a) Estimate the **temperature** of the surface of our **sun**, given that the Sun emits light whose peak intensity occurs in the visible spectrum at around **500nm**. **(b)** Determine the **temperature** of the surface of our **earth** .

Solution: (a) From Wien's law

$$T_{\text{Sun}} = \frac{b}{\lambda_p} = \frac{2.90 \times 10^{-3}}{500 \times 10^{-9}} = 6000 \text{K}$$

$$b = 2.90 \times 10^{-3} \text{ m} \cdot \text{K}$$

$$M(T) = \sigma T^4$$

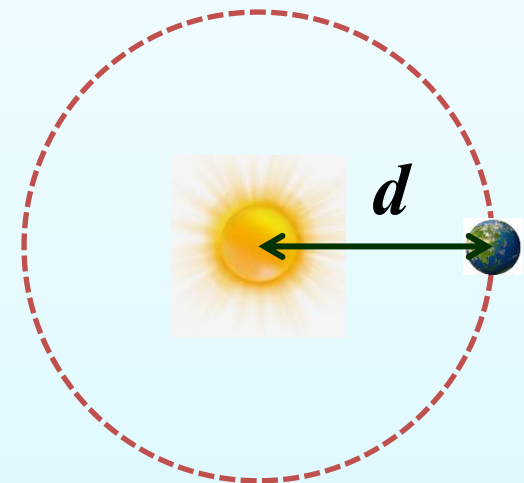
$$T \lambda_p = b$$

(b) Stefan-Boltzmann's law

$$M_{\text{Sun}} = \sigma T_s^4 4\pi R_s^2$$

$$M_{\text{earth}} = M_{\text{Sun}} \cdot \frac{\pi R_E^2}{4\pi d^2} = \sigma T_E^4 \cdot 4\pi R_E^2$$

$$T_{\text{earth}} = \sqrt{\frac{R_s}{2d}} T_s = 288 \text{K}$$



22.2 Photon theory of light and the Photoelectric Effect

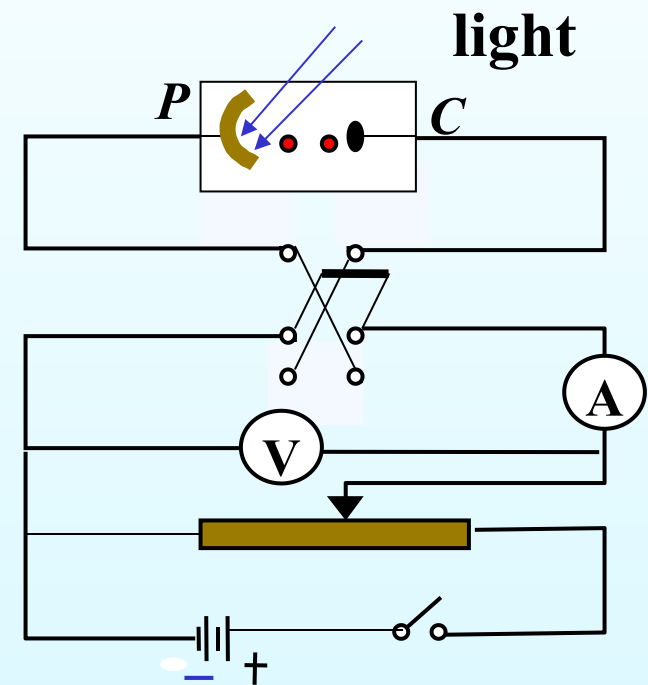
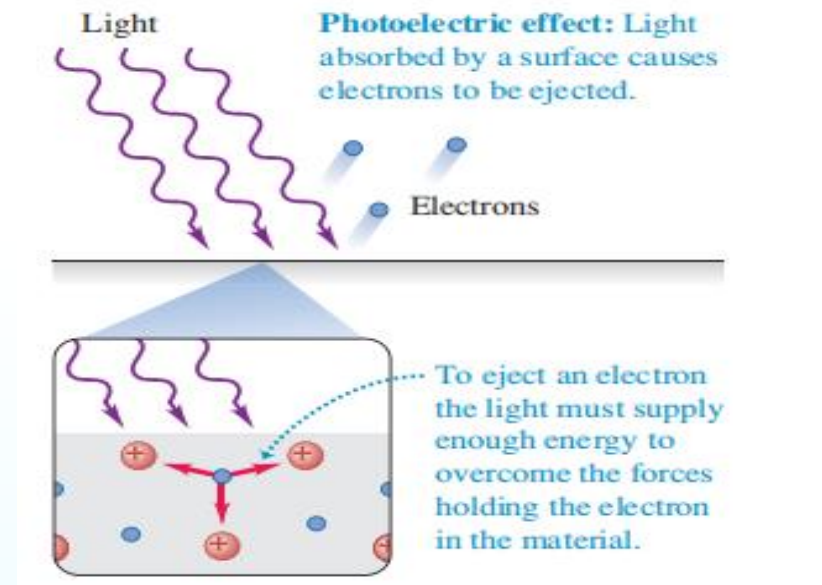
1. Photoelectric Effect

When light shines on a metal surface, **electrons** are emitted from the surface.

electrons (*photoelectrons*): 光电子
current (*photocurrent*): 光电流

P—a metal plate (阴极) emits photoelectrons with **various** kinetic energies.

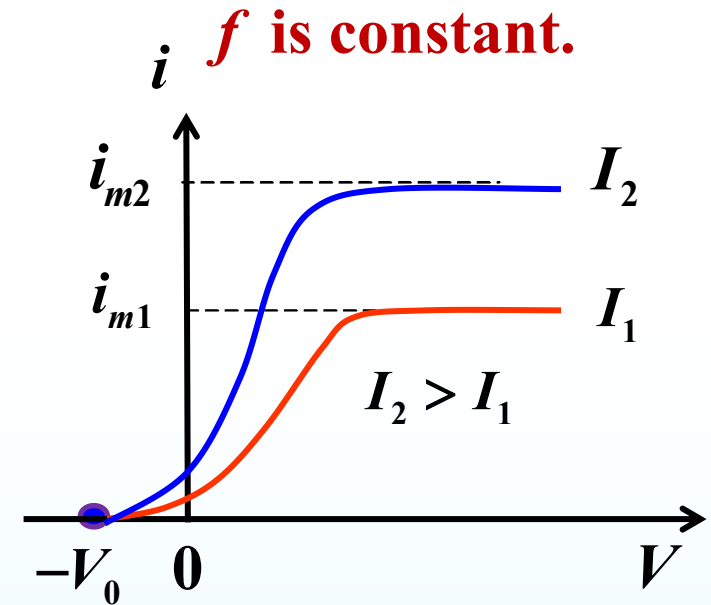
C—collector (阳极)



2. Law of experiment

(1) The saturation current i_m is proportional to the light intensity $i_m \propto I$.

(饱和光电流与入射光强度成正比)



(2) If the current reaches **zero**, reversed voltage V_0 should be connected.

(若使光电流为0, 需加反向电压遏止电压 V_0 .)

$$K_{\max} = eV_0$$

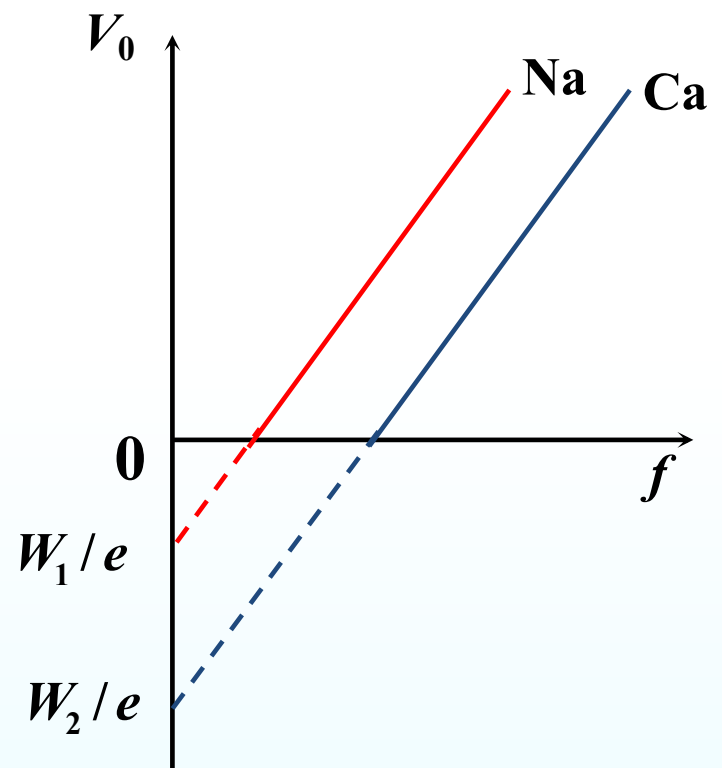
V_0 : stopping potential or stopping voltage 遏止电压

$K_{\max}$: the maximum kinetic energy 最大初动能

The **maximum kinetic energy** of ejected electrons is increasing with the **frequency** of the light and is independent of the intensity of light.

(光电子最大初动能随入射光的频率增加而线性增加，而与光的强度无关.)

$$V_0 = hf - \text{const}$$



(3) If $f < f_0$, no electrons will be ejected at all.

f_0 : **threshold frequency** or cutoff frequency

截止频率或
红限频率

(4) No **time delay**

(光电效应是瞬间发生的, 时间不超过 10^{-9}s .)

From classical wave theory

$$\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B} \quad (\text{Poynting vector in vacuum})$$

(1) **No** threshold frequency.

The **intensity** of an electromagnetic wave depends on its **amplitude** but not **on** its **frequency**. So the photoelectric effect should occur for light of any frequency.

(2) The **maximum kinetic energy** should be increased with the light **intensity** and independent of the **frequency**.

(3) An electron should **take time** to absorb enough energy and there should be **time delay**.

If the light falling on the surface is very faint, some time may elapse before the total energy absorbed by the surface equals the work function.

3. Einstein's photoelectric effect equation

(1) Einstein's Photon Explanation

爱因斯坦光量子假说(1905)

(a) A beam of light consists of small packages of energy called **photons** or quanta. Photons travel at the **speed of light** and have **zero rest mass**.

一束光：是一束以光速运动的粒子流，这些粒子称为**光子**（或称**光量子**）

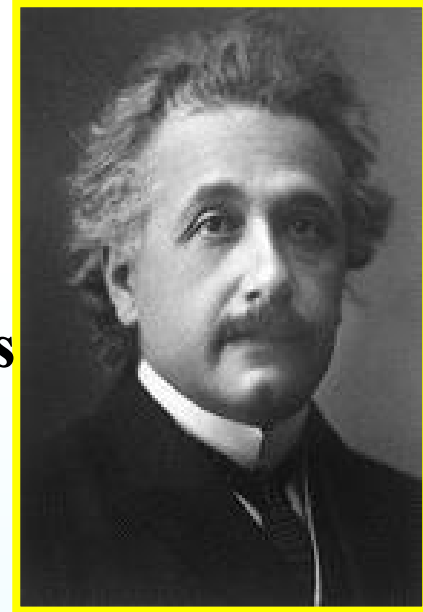
(b) The energy E of an individual photon is equal to a constant h times the photon frequency f .

$$E = h f$$

(energy of a photon)

(c) The **intensity** is proportional to the **number** of photons N across unit area in one second. $I = N h f$

光的强度决定于单位时间内通过单位面积的光子数



(2) Einstein's photoelectric effect equation

The electron can escape from the surface only if the energy it acquires is greater than the work function . As soon as photons of sufficient energy strike the surface, electrons can absorb them and be ejected.

当频率为 f 光照射金属时，一个电子是瞬间接收一个光子全部能量，若获得能量足以挣脱金属束缚，那么电子就可以释放出来，实现光电效应。

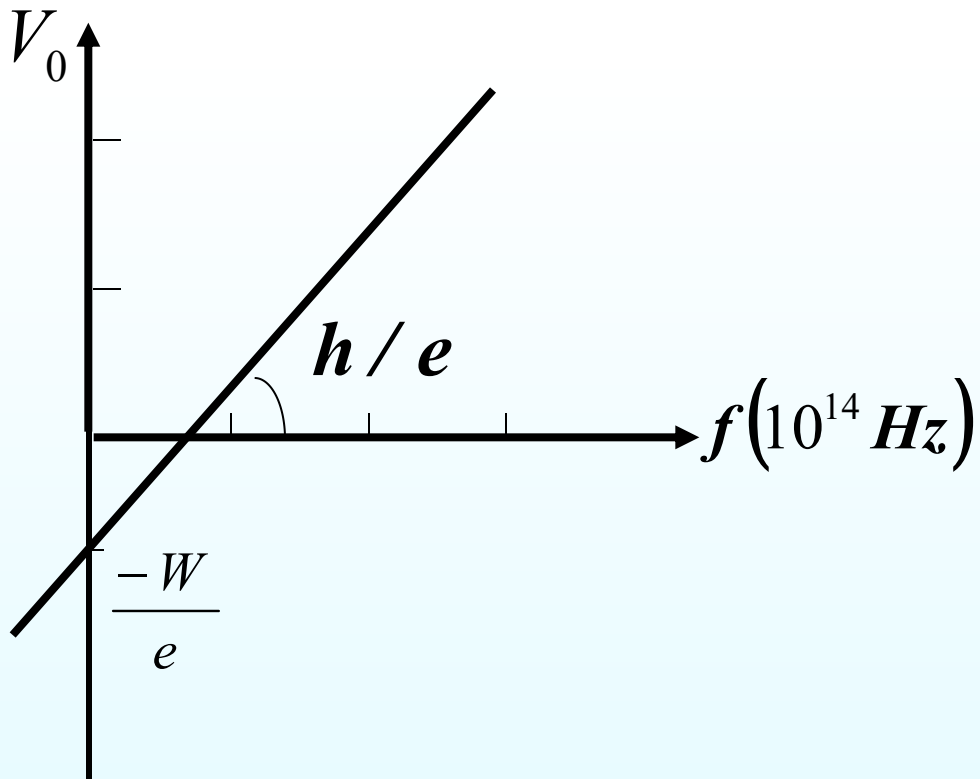
Conservation of energy

$$K_{\max} = h f - W$$

W : work function(逸出功，功函数)

Element	Work Function (eV)
Aluminum	4.3
Carbon	5.0
Copper	4.7
Gold	5.1
Nickel	5.1
Silicon	4.8
Silver	4.3
Sodium	2.7

photoelectric effect equation



Obtain Planck constant

$$h = \frac{\Delta V_0}{\Delta f} e$$

$$K_{\max} = h f - W$$

Stopping voltage

$$e V_0 = K_{\max}$$

Threshold frequency

$$f_0 = \frac{W}{h}$$

$$V_0 = \frac{h}{e} f - \frac{W}{e}$$

4. Light has the **wave-particle duality**

Light has the property of **wave** and at the same time has property of **particle** with speed **c** and **zero** rest mass.

photon: $m_0 = 0$

As a wave : f, λ

As a particle : m, p

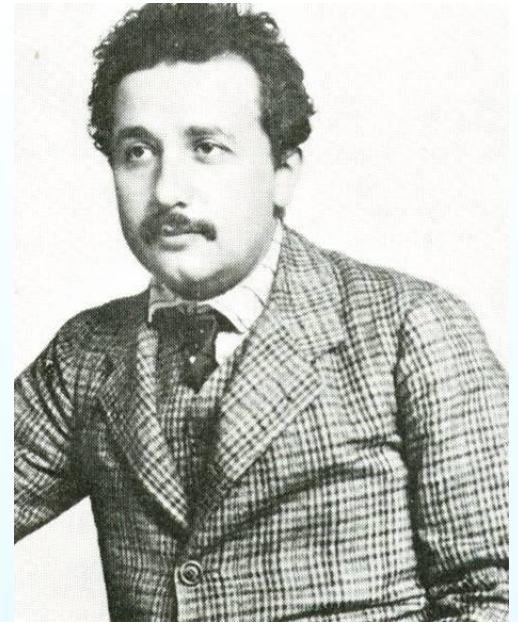
energy

$$E = hf = mc^2$$

momentum

$$p = mc = \frac{hf}{c^2} c = \frac{hf}{c}$$

$$p = \frac{h}{\lambda}$$



(a)



(b)



In a photoelectric effect experiment at a frequency above cut off, the **number of electrons** ejected is proportional to:

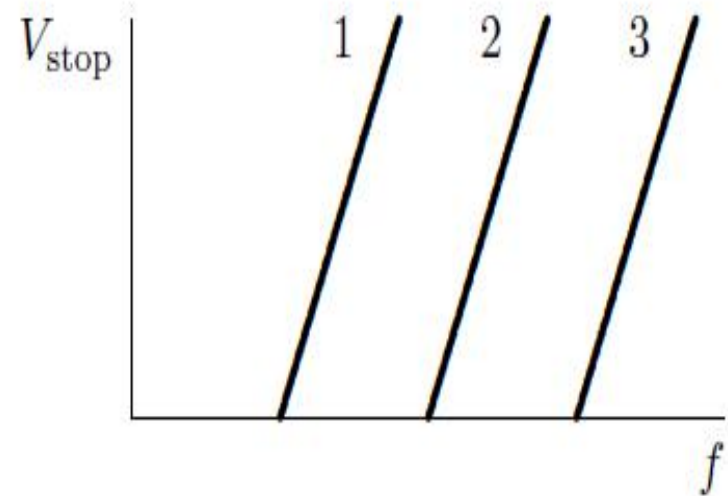


- A. their kinetic energy
- B. their potential energy
- C. the work function
- D. the frequency of the incident light
- ☒ E. the number of photons that hit the sample



The diagram shows the graphs of the stopping potential as a function of the frequency of the incident light for photoelectric experiments performed on three different materials. Rank the materials according to the values of their **work functions**, from least to greatest.

- A.** 1, 2, 3
- B. 3, 2, 1
- C. 2, 3, 1
- D. 2, 1, 3
- E. 1, 3, 2



Example Photons from a lightbulb. Estimate **how many** visible light photons a 100-W lightbulb emits per second.

Solution: Assume an average wavelength of the visible spectrum $\lambda = 500 \text{ nm}$.

$$E = Nhf$$

$$N = \frac{E}{hf} = \frac{E\lambda}{hc} = \frac{100 \times 500 \times 10^{-9}}{6.63 \times 10^{-34} \times 3.0 \times 10^8} = 2.5 \times 10^{20}$$

Example The **threshold wavelength** for the emission of photoelectrons from calcium is 384nm.

Determine the **work function** of calcium.

Solution:

$$W = hf - K_{\max} = hf_0 = h \frac{c}{\lambda_{\text{cutoff}}}$$

$$= 6.63 \times 10^{-34} \times \frac{3 \times 10^8}{384 \times 10^{-9}} = 5.18 \times 10^{-19} \text{ J (3.24 eV)}$$

Example The stopping potential for electron emitted from a surface of a metal illuminated by light of wavelength 491nm is 0.71V. When the incident wavelength is changed to a new value, the stopping potential is 1.43V. (a) What is new **wavelength**? (b) What is the **work function** for this metal?

Solution: $K_{\max} = hf - W$ $eV_0 = K_{\max}$ $f_0 = \frac{W}{h}$

$$V_0 = \frac{h}{e} f - \frac{W}{e} \quad V_1 = \frac{hc}{e\lambda_1} - \frac{W}{e} \quad V_2 = \frac{hc}{e\lambda_2} - \frac{W}{e}$$

$$\lambda_2 = \frac{hc}{e(V_2 - V_1) + \frac{hc}{\lambda_1}} \Rightarrow \lambda_2 = 382 \text{ nm}$$

$$W = h \frac{c}{\lambda_1} - K_{1\max} = h \frac{c}{\lambda_1} - eV_1 \Rightarrow W = 2.91 \times 10^{-19} \text{ J}$$

Example: Determine the **energy** and **momentum** of photons with follow wavelength: 15000Å infrared light , 200Å ultraviolet light, 0.01Å γ -ray.

$$P = \frac{h}{\lambda} \quad E = \frac{hc}{\lambda} = Pc$$

$$P_1 = \frac{h}{\lambda_1} = \frac{6.626 \times 10^{-34}}{15000 \times 10^{-10}} = 4.42 \times 10^{-28} \text{ kg} \cdot \text{m} / \text{s} \quad E_1 = P_1 c = 1.33 \times 10^{-19} \text{ J}$$

$$P_2 = \frac{h}{\lambda_2} = \frac{6.626 \times 10^{-34}}{200 \times 10^{-10}} = 3.31 \times 10^{-26} \text{ kg} \cdot \text{m} / \text{s} \quad E_2 = P_2 c = 9.34 \times 10^{-18} \text{ J}$$

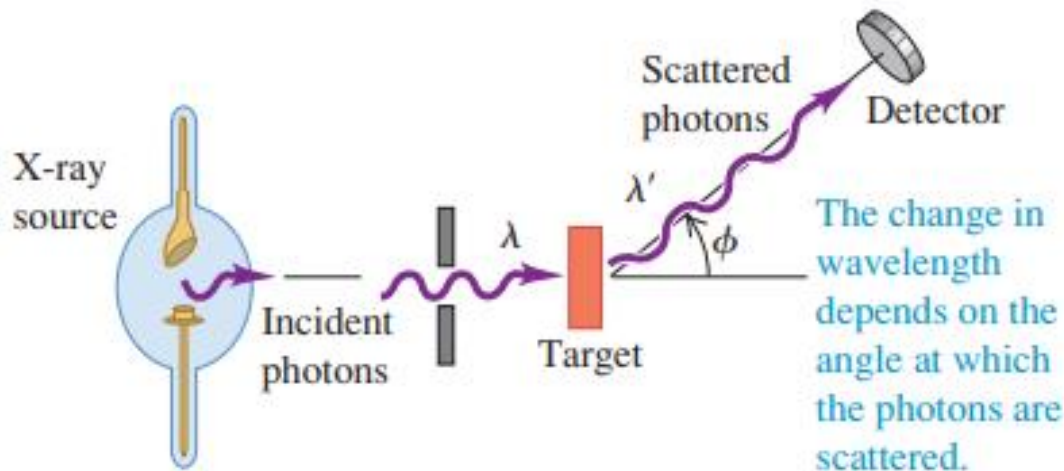
$$P_3 = \frac{h}{\lambda_3} = \frac{6.626 \times 10^{-34}}{0.01 \times 10^{-10}} = 6.63 \times 10^{-22} \text{ kg} \cdot \text{m} / \text{s} \quad E_3 = P_3 c = 1.99 \times 10^{-13} \text{ J}$$

Which are true and which are false?

- 1. The greater the **frequency** of the incident light is, the greater is the **stopping potential**.**
- 2. The greater the **intensity** of the incident light is, the greater is the **cutoff frequency**.**
- 3. The greater the **work function** of the target material is, the greater is the **stopping potential**.**
- 4. The greater the **work function** of the target material is, the greater is the **cutoff frequency**.**
- 5. The greater the **light frequency** is, the greater is the **maximum kinetic energy** of the ejected electrons.**
- 6. The greater the **energy** of the photons is, the smaller is the **stopping potential**.**

22.3 Compton scattering (1923)

Further evidence of the correctness of the photon concept by A.H. Compton ,who measured the scattering of X-rays by free electrons.

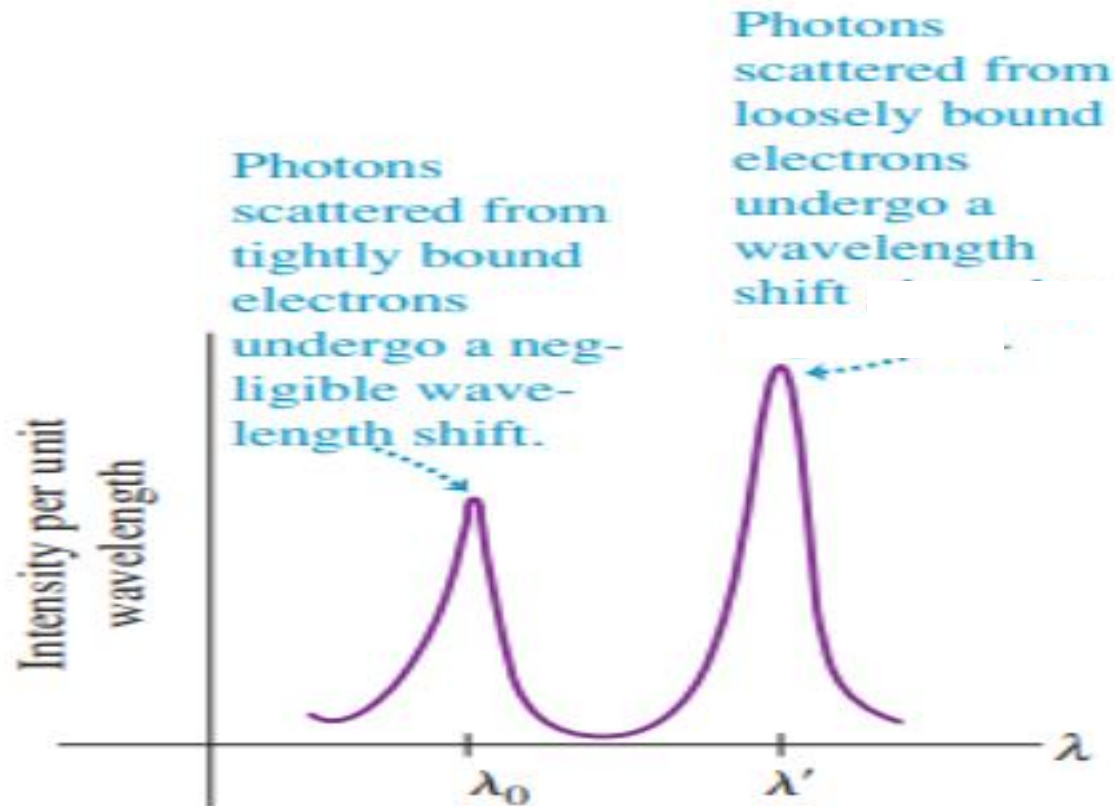


He found that the scattered light had a slightly **longer wavelength** than that of the incident light, and therefore a slightly lower frequency indicating a loss of energy.

λ_0 : 正常光
 λ ($> \lambda_0$) 康普顿散射光

$$\Delta\lambda = \lambda - \lambda_0 = 0.048 \sin^2 \frac{\phi}{2} \text{ (}\overset{\circ}{\text{A}}\text{)}$$

Law of experiment



Some of the scattered radiation has smaller frequency (**longer wavelength**) than the incident radiation and that the **change in wavelength** depends on the **angle** through which the radiation is scattered.

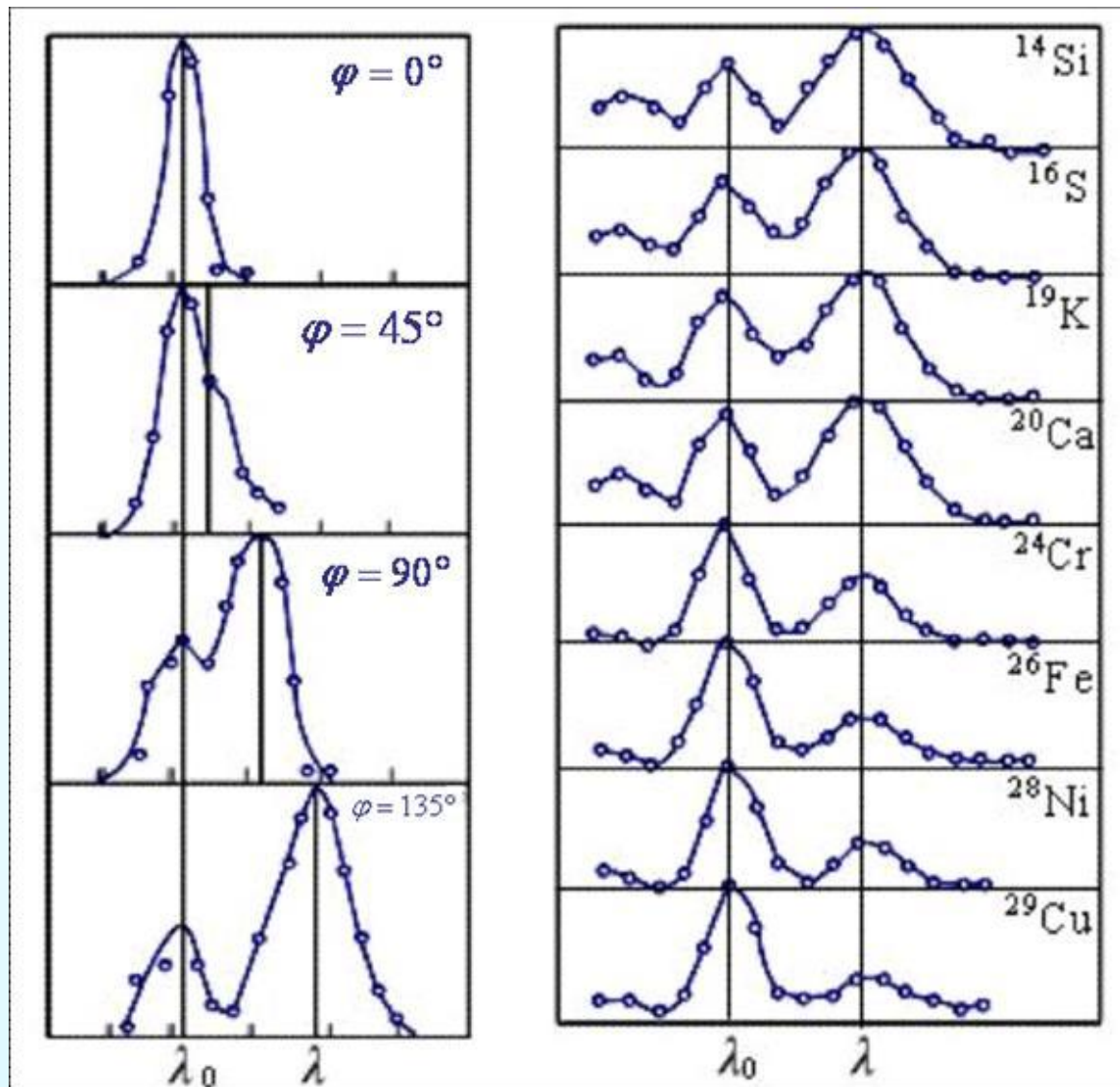
Law of experiment

$$\Delta\lambda = \frac{h}{m_0 c} (1 - \cos\varphi)$$

1. 在原子量小的物质中，康普顿散射较强，反之较弱。

2. 波长的改变量 $\Delta\lambda = \lambda - \lambda_0$ 随散射角 φ 的增加而增加。

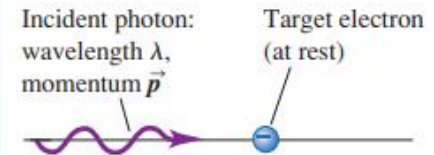
3. 对不同的散射物质，只要在同一散射角下，波长的改变量 $\Delta\lambda = \lambda - \lambda_0$ 都相同。



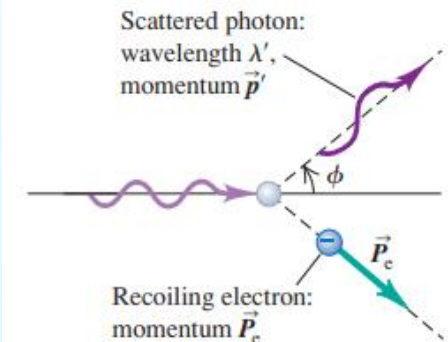
Wave-Model Prediction: In the wave description, scattering would be a process of **absorption and re-radiation**. Part of the energy of the light wave would be absorbed by the electron, which would oscillate in response to the oscillating electric field of the wave. The oscillating electron would act like a miniature antenna, re-radiating its acquired energy as scattered waves in a variety of directions. The frequency at which the electron oscillates would be the **same** as the frequency of the incident light, and the re-radiated light would have the same frequency (same wavelength) as the oscillations of the electron.

Photon-Model Prediction: In the photon model we imagine the scattering process as **a collision** of two particles, the incident photon and an electron that is initially at rest. The incident photon would **give up part of its energy and momentum to the electron**, which recoils as a result of this impact. The scattered photon that remains can fly off at a variety of angles with respect to the incident direction, but it has less energy and less momentum than the incident photon. Therefore, in the photon model, the scattered light has a **lower frequency** and **longer wavelength** than the incident light.

(a) Before collision: The target electron is at rest.



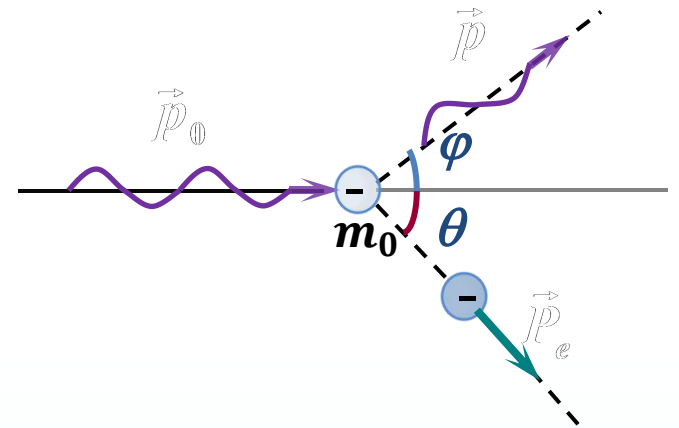
(b) After collision: The angle between the directions of the scattered photon and the incident photon is ϕ .



Conservation of energy

$$p_0 c + m_0 c^2 = p c + E_e \quad \text{--- (1)}$$

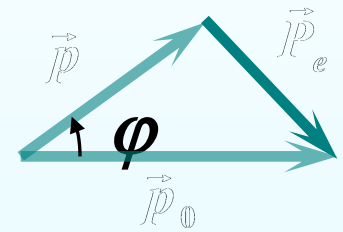
$$(p_0 c - p c + m_0 c^2)^2 = E_e^2 = (m_0 c^2)^2 + (P_e c)^2 \quad \text{--- (2)}$$



Conservation of momentum

$$\vec{P}_e = \vec{p} - \vec{p}_0$$

$$P_e^2 = p^2 + p_0^2 - 2 p p_0 \cos \phi \quad \text{--- (3)}$$

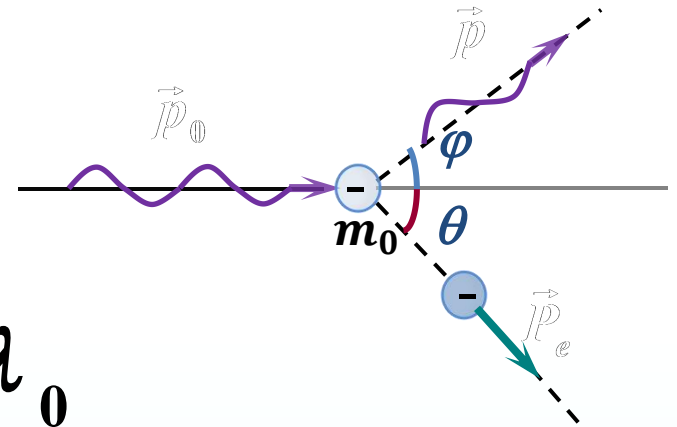


Substitute (3) into (2), and divided by $p p_0$

$$\frac{m_0 c}{p} - \frac{m_0 c}{p_0} = 1 - \cos \phi \quad \text{--- (4)}$$

$$\frac{m_0 c}{p} - \frac{m_0 c}{p_0} = 1 - \cos \varphi \quad (4)$$

$$p_0 = \frac{h}{\lambda_0}, \quad p = \frac{h}{\lambda} \quad \Delta\lambda = \lambda - \lambda_0$$



$$\Delta\lambda = \frac{h}{m_0 c} (1 - \cos \varphi) = 2\lambda_c \sin^2 \frac{\varphi}{2}$$

$\Delta\lambda$: Compton shift

λ_c : Compton wavelength for **an electron**

$$\lambda_c = \frac{h}{m_0 c} \approx 2.43 \times 10^{-3} \text{ nm} = \mathbf{2.3 \text{ pm}}$$

$$\varphi = 90^\circ, \quad \Delta\lambda = \lambda_c; \quad \varphi = 180^\circ, \quad \Delta\lambda = 2\lambda_c$$

Calculation of the **recoiling electron**

$$(1) E_K = E - E_0 = h\nu_0 - h\nu$$

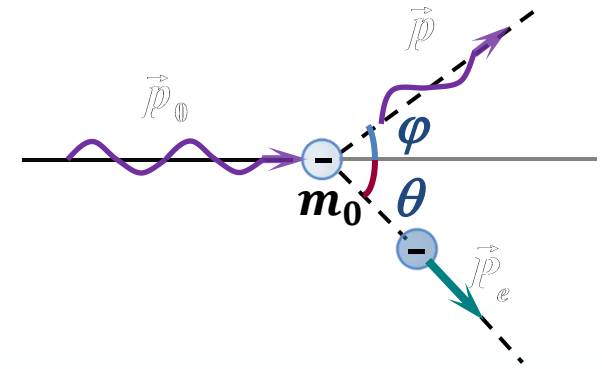
(2) Determine momentum

$$(a) E_K = E - E_0 = \frac{m_0 c^2}{\sqrt{1 - v^2/c^2}} - m_0 c^2 \Rightarrow v \rightarrow P_e = \frac{m_0}{\sqrt{1 - v^2/c^2}} v$$

$$(b) P_e = \frac{1}{c} \sqrt{E^2 - E_0^2} = \frac{1}{c} \sqrt{(E + E_0)(E - E_0)} = \frac{1}{c} \sqrt{E_K (E_K + 2E_0)}$$

Direction of momentum

$$\frac{h}{\lambda} \sin \varphi - P_e \sin \theta = 0 \Rightarrow \theta = \arcsin\left(\frac{h}{P_e \lambda} \sin \varphi\right)$$



The Compton effect adds to the firm foundation for the photon theory of light.

进一步确认了光的粒子性；

The law of Conservation of energy and momentum is still available for microscopic particles.

确认了动量守恒定律与能量守恒定律在微观粒子相互作用中的正确性。

讨论:

$$\Delta\lambda = \frac{h}{m_0 c} (1 - \cos\varphi) = 2\lambda_c \sin^2 \frac{\varphi}{2}$$

- 入射光子与散射物质中**束缚微弱**的电子弹性碰撞时，一部分能量传给电子，散射光子能量减少，波长变大.
- $\Delta\lambda$ 与 φ 有关而**与物质无关**，是光子与近自由电子间的相互作用.
- 散射中 $\Delta\lambda = 0$ 的散射光是因**光子与紧束缚电子**的作用. 原子量大的物质，其电子束缚较强，因而康普顿效应不明显.
- 若 $\lambda_0 \gg \lambda$ 则 $\lambda \approx \lambda_0$ ，**可见光**观察**不到**康普顿效应.
- 光具有**波粒二象性**：光在传递过程中，波动性较为显著；光与物质相互作用时，粒子性比较显著.

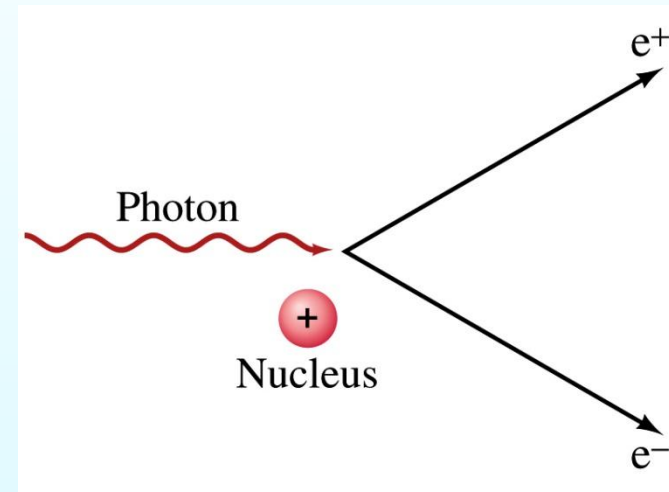
Different Effect of Photon Interactions

1. The photon can be scattered from an electron (or a nucleus) and in the process lose some energy and its f will be lower —— *Compton Effect* (当光子与原子中的自由电子或弱束缚电子弹性碰撞时发生康普顿效应).
2. A photon may knock an electron out of an atom and in the process itself disappear—— *photoelectric* (光子与原子中束缚电子发生完全非弹性碰撞时, 光子被电子吸收, 发生光电效应)

3. A photon may knock an atomic electron to a higher energy state in the atom in the process photon disappears, and all its energy is given to the atom. (Such an atom is in an **excited state**).

4. A photon can actually create matter, such as the production of an electron and a positron. This process is called *pair production* (正负电子偶).

光与物质的相互作用时可能会发生若干种不同的效应: 有光电效应, 康普顿效应, 还产生正负电子偶, 它们是不同的能量的光子与物质中的原子、电子、原子核相互作用的结果。



In Compton scattering from stationary particles **the maximum change in wavelength** can be made smaller by using:



- A. higher frequency radiation
- B. lower frequency radiation
- ☒ C. more massive particles
- D. less massive particles
- E. particles with greater charge

Example X-ray scattering. X-rays of wavelength 0.140nm are scattered from a block of carbon. What will be the **wavelengths** of X-rays scattered at (a) 0°, (b) 90°, (c) 180°?

Solution:
$$\Delta\lambda = \frac{h}{m_0 c} (1 - \cos \varphi)$$

(a)
$$\Delta\lambda = \frac{h}{m_0 c} (1 - \cos \varphi) = 0 \quad \lambda = \lambda_0 + \Delta\lambda = 0.140\text{nm}$$

(b)
$$\Delta\lambda = \frac{h}{m_0 c} (1 - \cos \varphi) = \lambda_c = 2.43\text{pm}$$

$$\lambda = \lambda_0 + \lambda_c = 0.140\text{nm} + 2.4 \times 10^{-12} \text{m} = 0.142\text{nm}$$

(c)
$$\Delta\lambda = \frac{h}{m_0 c} (1 - \cos \varphi) = 2\lambda_c = 4.86\text{pm}$$

$$\lambda = \lambda_0 + 2\lambda_c = 0.140\text{nm} + 4.9 \times 10^{-12} \text{m} = 0.145\text{nm}$$

Example X-rays with $\lambda_0=100\text{pm}$ are scattered from a carbon target. The scattered radiation is viewed at 90° to the incident beam. (a). What is a Compton shift $\Delta\lambda$? (b). What **kinetic energy** is imparted to the recoiling electron?

Solution:

$$\Delta\lambda = \frac{h}{m_0 c} (1 - \cos \varphi) = \frac{h}{m_0 c} = 2.43\text{pm}$$

$$K_e = hf_0 - hf = \frac{hc}{\lambda_0} - \frac{hc}{\lambda_0 + \Delta\lambda}$$

$$K_e = \frac{hc\Delta\lambda}{\lambda_0 (\lambda_0 + \Delta\lambda)} = 295\text{eV}$$

Example X-ray of wavelength $\lambda=22\text{pm}$ (photon energy =56 keV) are scattered from a carbon target, and the scattered rays are detected at 85° to the incident beam.

(a) What is the Compton shift of the scattered rays?

(b) What percentage of the initial X-ray photo energy is transferred to an electron in such scattering?

Solution: (a)
$$\Delta\lambda = \frac{h}{m_0 c}(1 - \cos\varphi)$$
$$= \frac{(6.63 \times 10^{-34})(1 - \cos 85^\circ)}{(9.11 \times 10^{-31})(3.0 \times 10^8)} = 2.21 \times 10^{-12} \text{ m} \approx 2.2 \text{ pm}$$

(b)
$$\text{frac} = \frac{\Delta E}{E_{\text{initial}}} = \frac{hf_0 - hf}{hf_0} = \frac{\lambda - \lambda_0}{\lambda}$$

$$\text{frac} = \frac{\Delta\lambda}{\lambda_0 + \Delta\lambda} = \frac{2.21}{22 + 2.21} = 0.091 (9.1\%)$$

例 在康普顿实验中，当能量为**0.50MeV**的X射线的光子射中一个静止电子时，该电子获得的动能为**0.20MeV**

求：(1) 散射光子的波长、能量、动量与质量。

(2) 散射光子与入射方向的夹角。

解：(1) 碰撞过程中能量守恒

$$hf_0 + m_0c^2 = hf + mc^2 \quad K_e = mc^2 - m_0c^2 = hf_0 - hf$$

散射光子的能量 $E = hf = hf_0 - K = 0.50 - 0.20 = 0.30\text{MeV}$

散射光子的波长 $E = hf = \frac{hc}{\lambda}$

$$\longrightarrow \lambda = \frac{hc}{E} = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{0.30 \times 10^6 \times 1.6 \times 10^{-19}} = 4.14 \times 10^{-12} \text{m}$$

散射光子的动量

$$p = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}}{4.14 \times 10^{-12}} = 1.60 \times 10^{-22} \text{ kg} \cdot \text{m} \cdot \text{s}^{-1}$$

散射光子的质量

$$m_{\varphi} = \frac{E}{c^2} = \frac{hf}{c^2} = \frac{p}{c}$$

$$= \frac{1.60 \times 10^{-22}}{3 \times 10^8} = 5.34 \times 10^{-31} \text{ kg}$$

(2) 由入射光子的能量可得入射光的波长为

$$\lambda_0 = \frac{hc}{E_0} = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{0.5 \times 10^6 \times 1.6 \times 10^{-19}} = 2.49 \times 10^{-12} \text{ m}$$

$$\Delta\lambda = \lambda - \lambda_0 = \lambda_c (1 - \cos\varphi) = 2\lambda_c \sin^2 \frac{\varphi}{2}$$

$$\sin \frac{\varphi}{2} = \left[\frac{(\lambda - \lambda_0)}{2\lambda_c} \right]^{\frac{1}{2}} \approx 0.5832 \quad \rightarrow \varphi = 71.36^\circ$$

Key words And Terms

spectrum (spectra)

光谱

spectral line

光谱线

stationary states

稳定态

quantum number

量子数

energy level or energy state

能级

ground state

基态

excited state

激发态

binding energy

束缚能

ionization energy

电离能

radioactivity

放射性

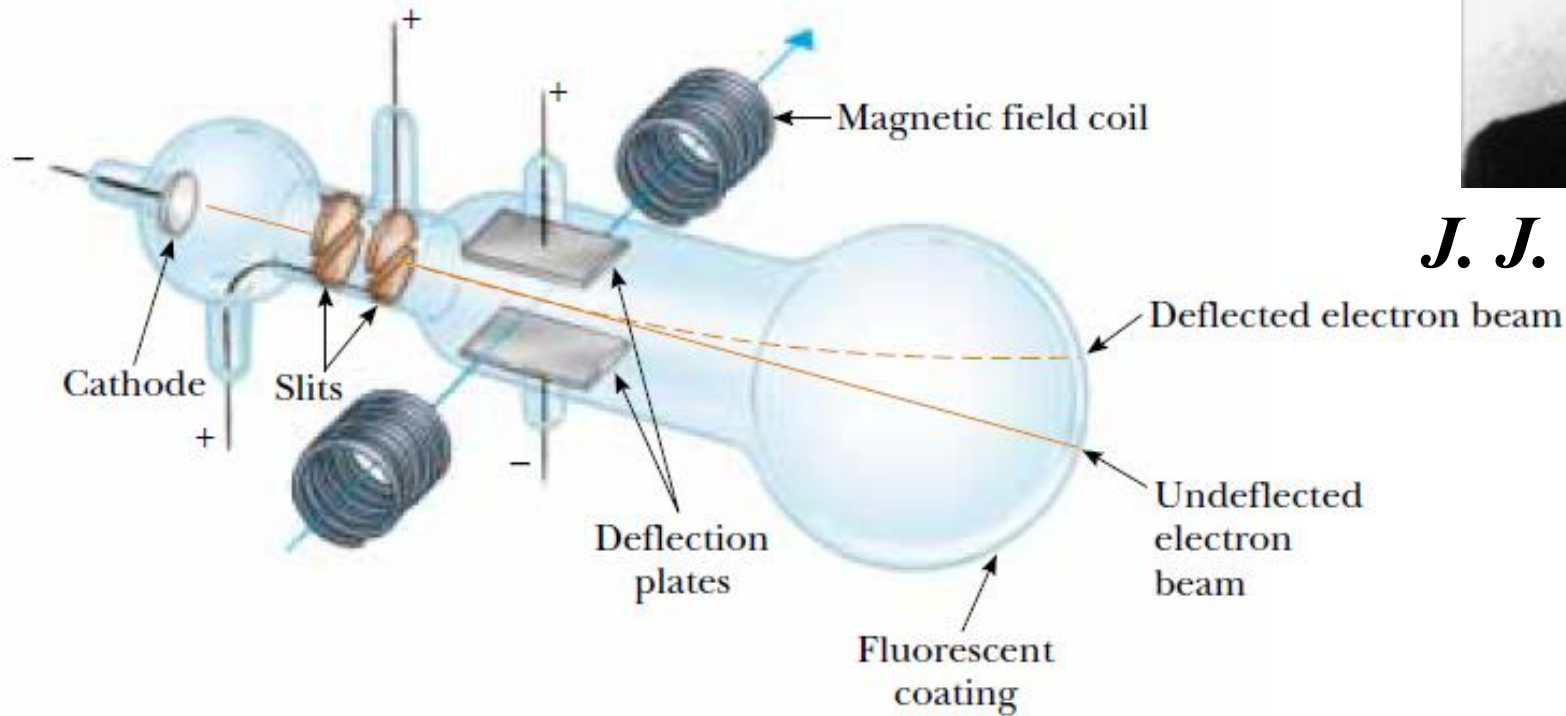
matter waves

物质波

Discovery of the Electron



J. J. Thomson



Thomson's apparatus for measuring e/m_e (charge to mass ratio)

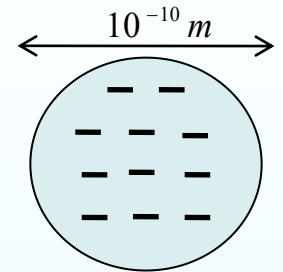
The credit for the discovery is generally given to Thomson. He believed that electron was due to *tiny negatively charged particles* and made careful measurements on them.

22.4 The Bohr model of hydrogen atom

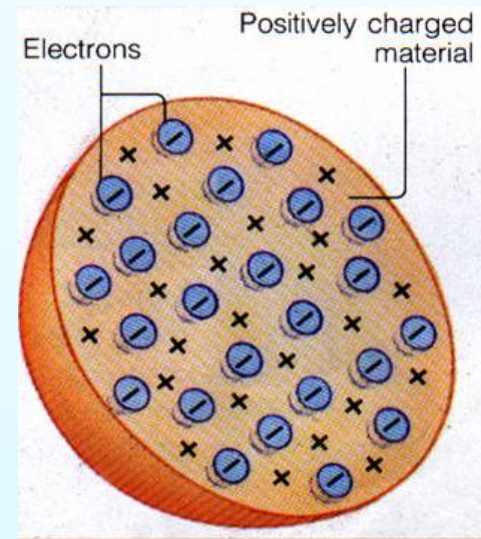
1. Early Models of the Atom

(a) In 1897, J.J Thomson found electrons.

In 1903, Thomson's model, “plum pudding model”
Tiny negatively charged electrons were inside of
a homogeneous sphere of positive charge.



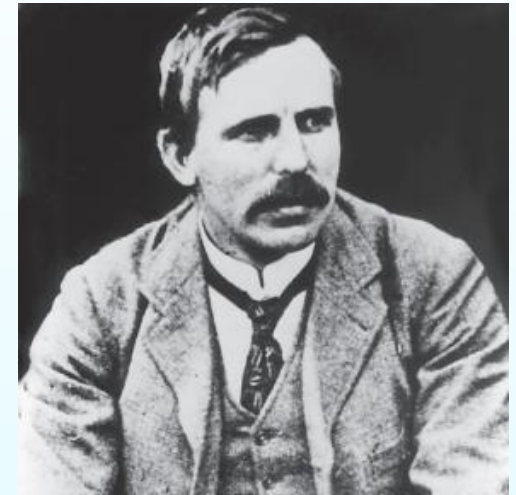
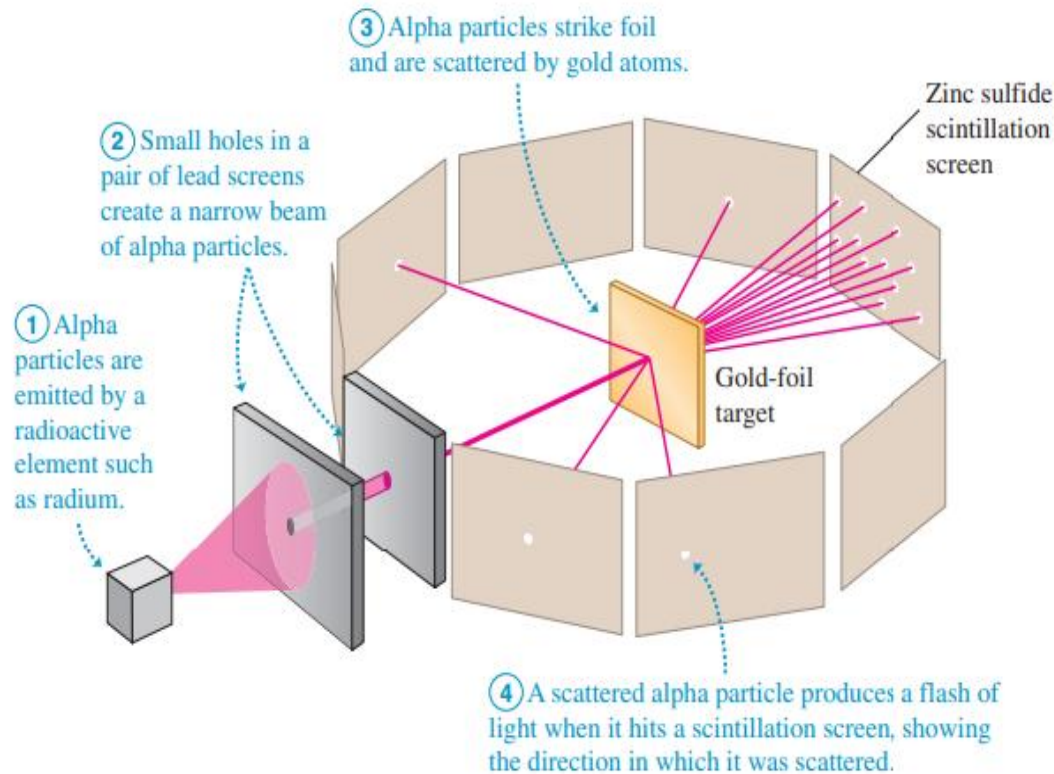
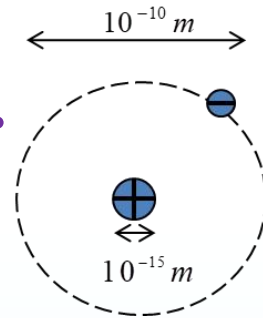
difficulty: unable to
explain why atoms emit
line spectra.



(b) In 1911, Ernest Rutherford's model *nuclear model* or *planetary model* (行星模型, 核式模型)

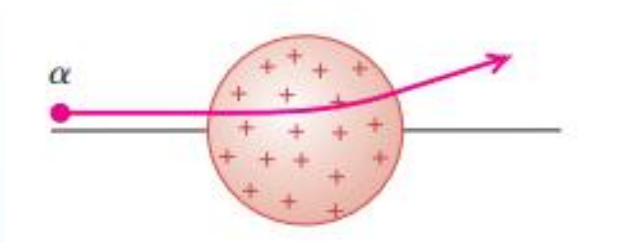
Electrons were moving in orbits about the nucleus.

The atom does have a nucleus—a very **small**, very **dense** structure.

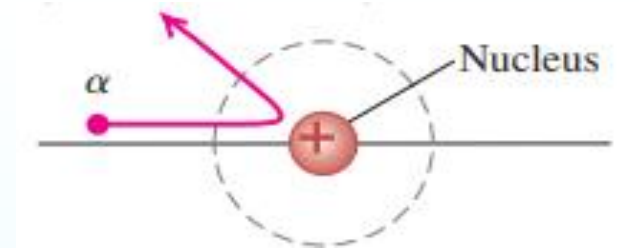


Ernest Rutherford
(1871–1937), 1908 “Nobel Prize in **chemistry**”

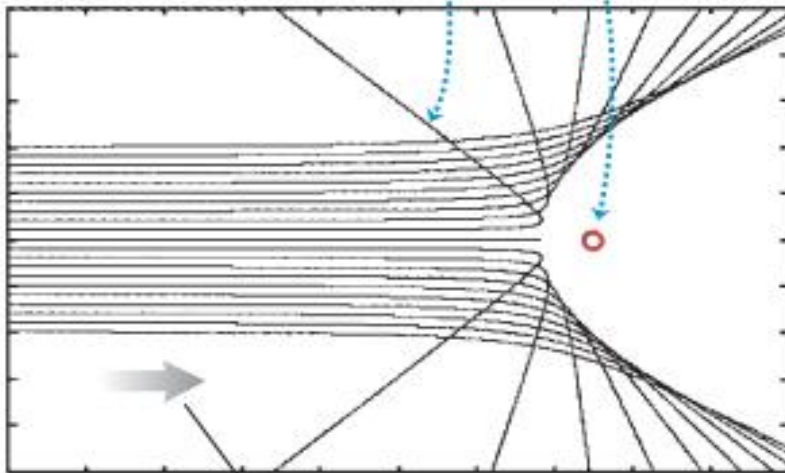
Thomson's model of the atom:
An alpha particle is scattered through only a small angle.



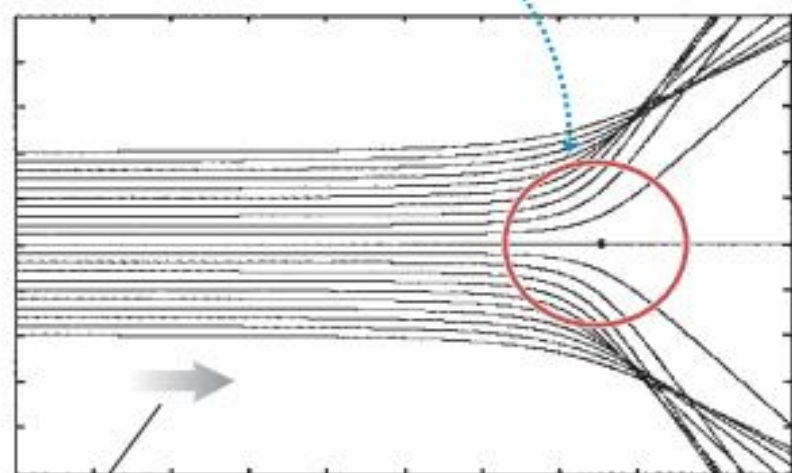
Rutherford's model of the atom: An alpha particle can be scattered through a large angle by the compact, positively charged nucleus.



(a) A gold nucleus with radius 7.0×10^{-15} m gives large-angle scattering.



(b) A nucleus with 10 times the radius of the nucleus in (a) shows *no* large-scale scattering.



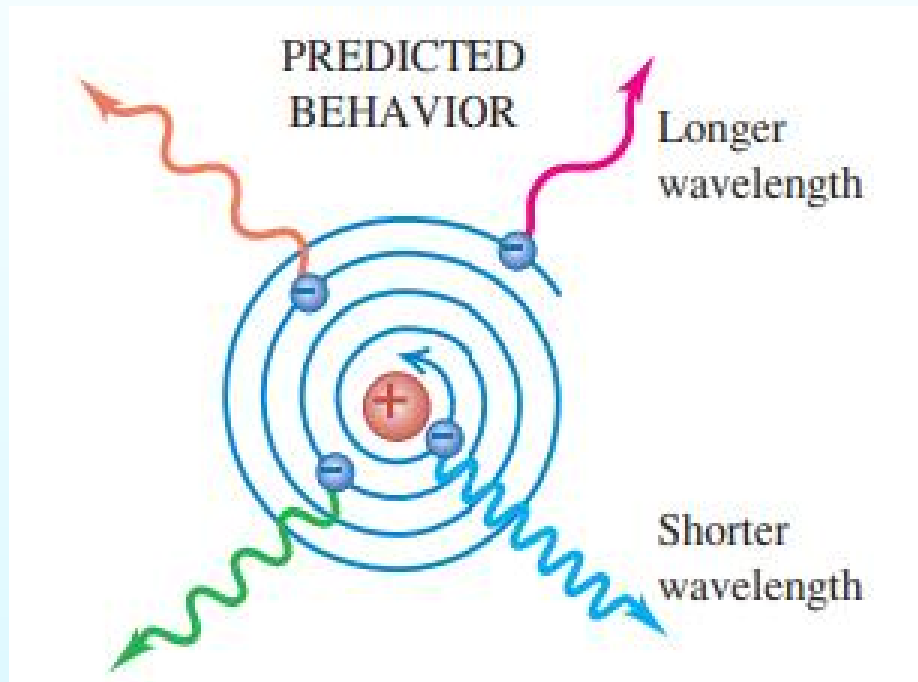
Motion of incident 5.0-MeV alpha particles

Difficulties:

(a) Unable to explain why atoms emit line spectra

According to **classical theory** the frequency of the electromagnetic waves emitted should **equal** the frequency of revolution. As the electrons radiated energy, their angular speeds would change continuously, and they would emit a continuous spectrum (a mixture of all frequencies), **not** the **line spectrum** actually observed.

(b) **Accelerating electrons** should lose their own energy and would be expected to spiral into the nucleus -atoms are **unstable**.



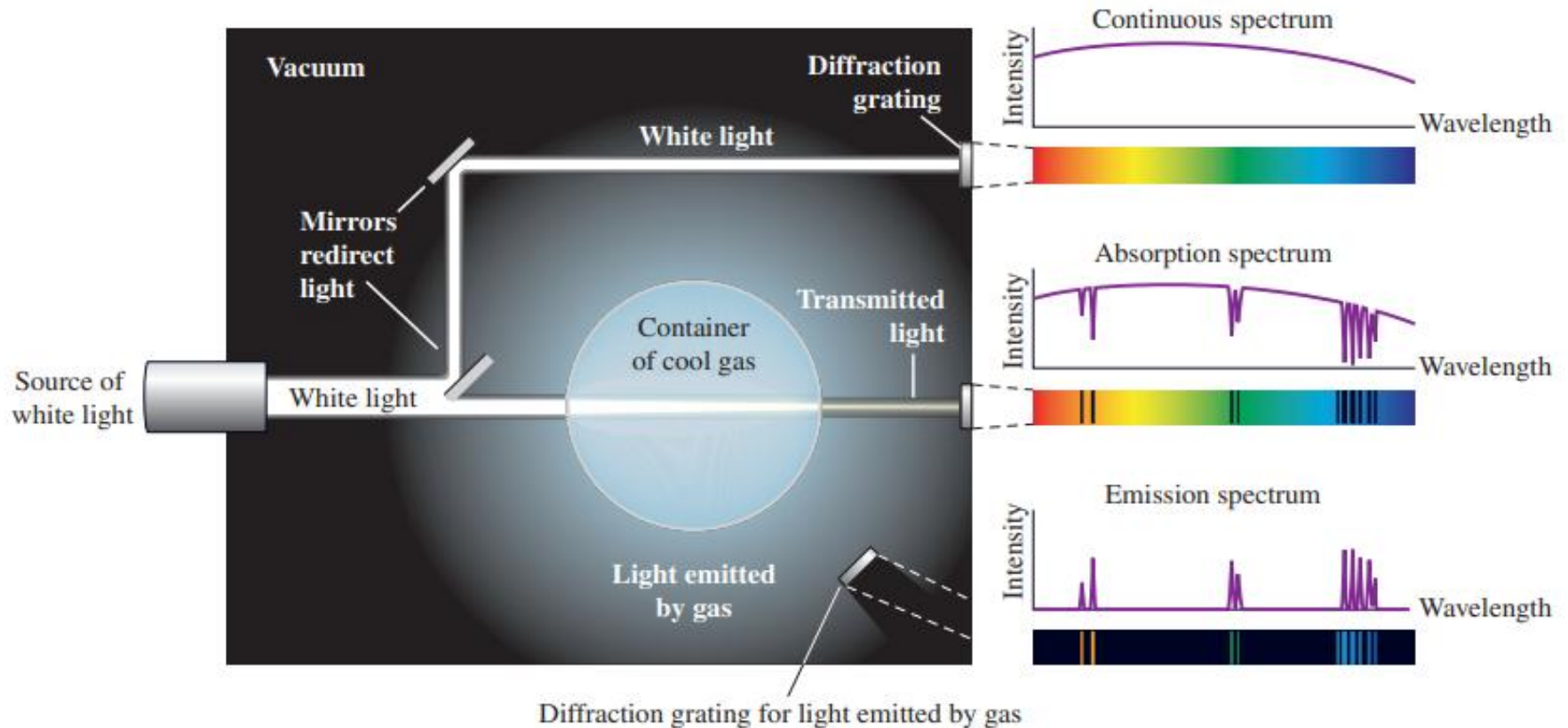
2. Spectrum of hydrogen

Continuous spectrum : solid , liquid

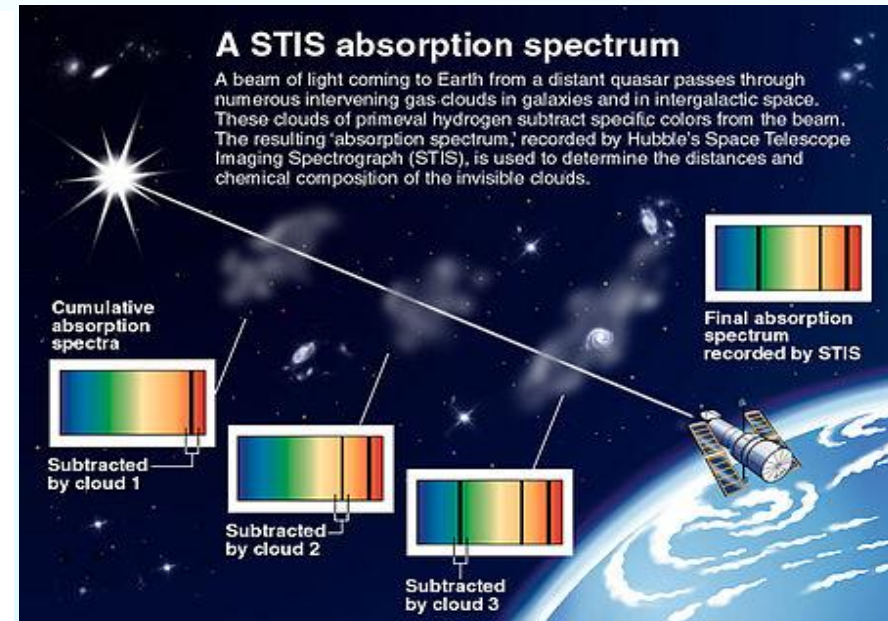
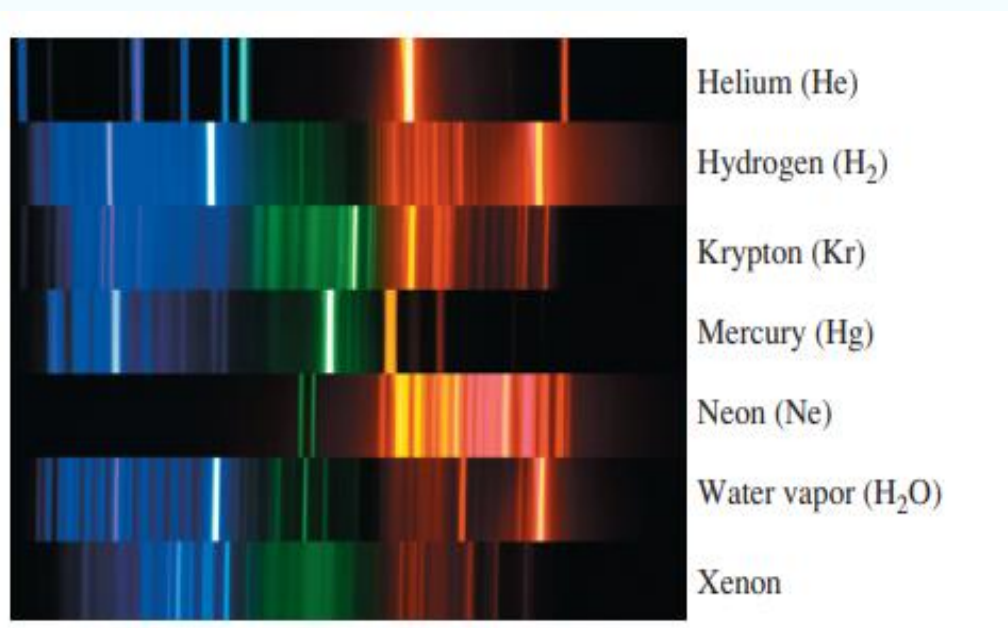
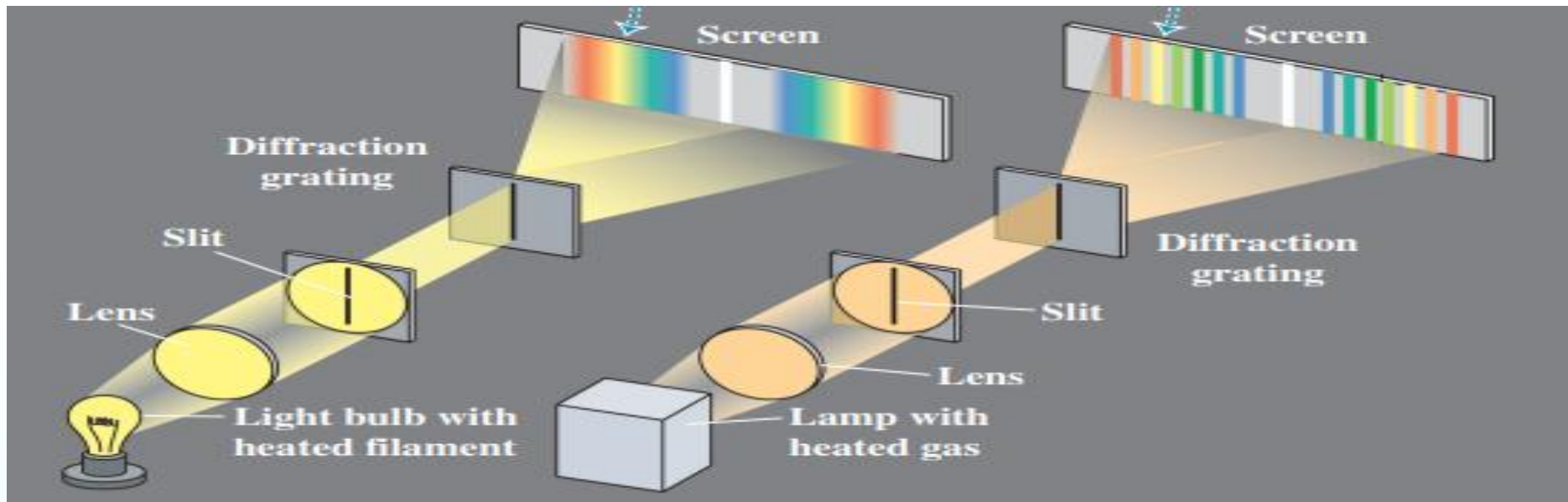
Line spectrum : gas, atoms

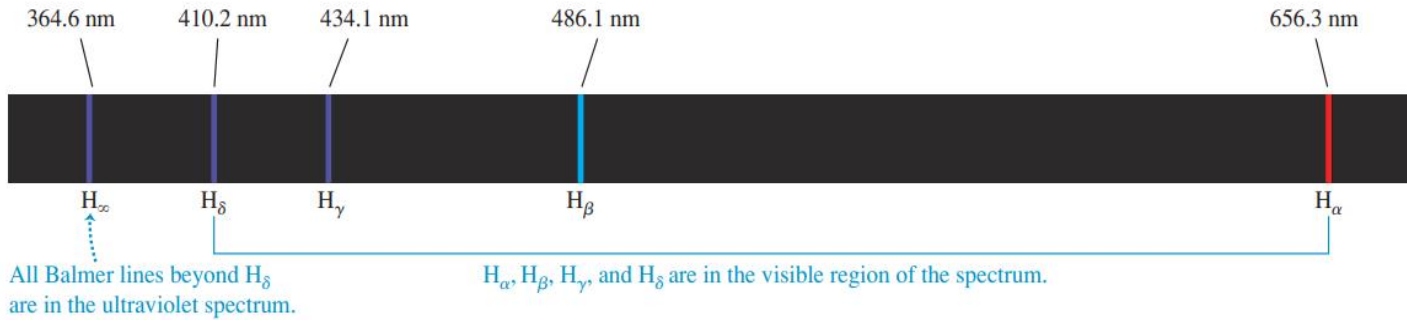
Emission spectrum: bright lines (fingerprint)

Absorption spectrum : dark lines



2. Spectrum of hydrogen





Johann Jakob Balmer
1825—1898

In 1885, J.J. Balmer showed an empirical formula

$$\lambda = B \frac{n^2}{n^2 - 2^2} \text{ nm}, \quad n = 3, 4, 5, \text{L} \quad B = 364.6 \text{ nm}$$

Rydberg rewrote the formula(1989)

$$\frac{1}{\lambda} = R \left(\frac{1}{m^2} - \frac{1}{n^2} \right), m = 1, 2, 3\text{L}$$

$$R = 1.0974 \times 10^7 \text{ m}^{-1} \quad \text{Rydberg constant}$$



Janne Rydberg
1854—1919

$$\frac{1}{\lambda} = R \left(\frac{1}{m^2} - \frac{1}{n^2} \right), m = 1, 2, 3, \dots$$

If m is fixed, $n = m+1, m+2, m+3$ up to infinity.

$m=1$, **Lyman** series (1914) (ultraviolet)

$$\frac{1}{\lambda} = R_H \left(\frac{1}{1^2} - \frac{1}{n^2} \right) \quad n = 2, 3, 4, \dots$$

$m=2$, **Balmer** series (visible)

$$\frac{1}{\lambda} = R_H \left(\frac{1}{2^2} - \frac{1}{n^2} \right) \quad n = 3, 4, 5, \dots$$

$m=3$, **Paschen** series (1908) (infrared)

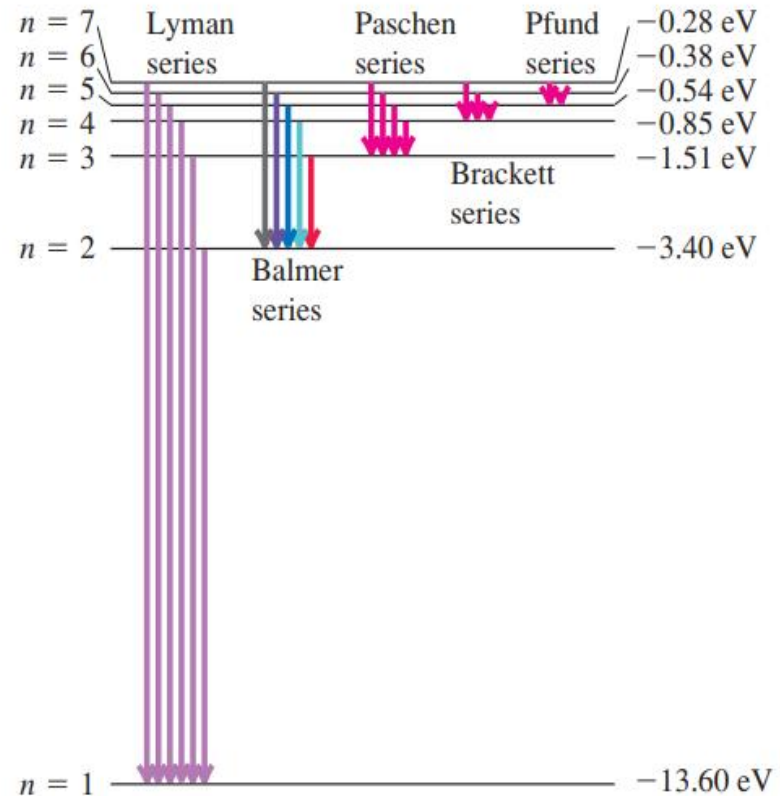
$$\frac{1}{\lambda} = R_H \left(\frac{1}{3^2} - \frac{1}{n^2} \right) \quad n = 4, 5, 6, \dots$$



T. Lyman
1874-1954



F. Paschen
1865-1947



3. Bohr's model

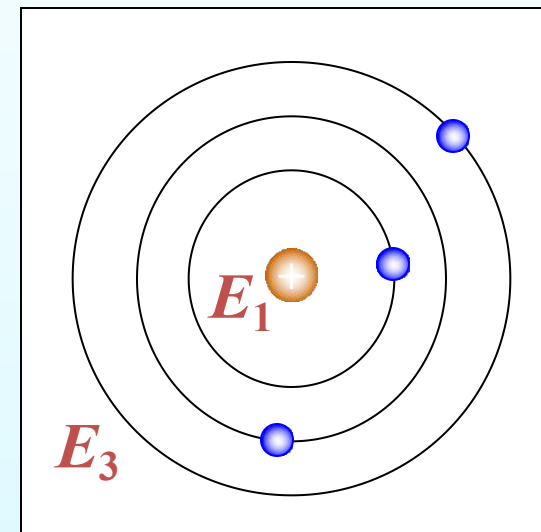
(1) Bohr's assumption about hydrogen model

(a) Stationary states(定态假设): there are several stable orbits r_1, r_2, r_3 on which radiation don't occur and energy has certain value E_1, E_2, E_3 .

电子在原子中可以在一些特定的圆轨道上运动而不辐射电磁波，这时，原子处于稳定状态，简称**定态**。



David BohrNiels
Bohr (1885–1962)

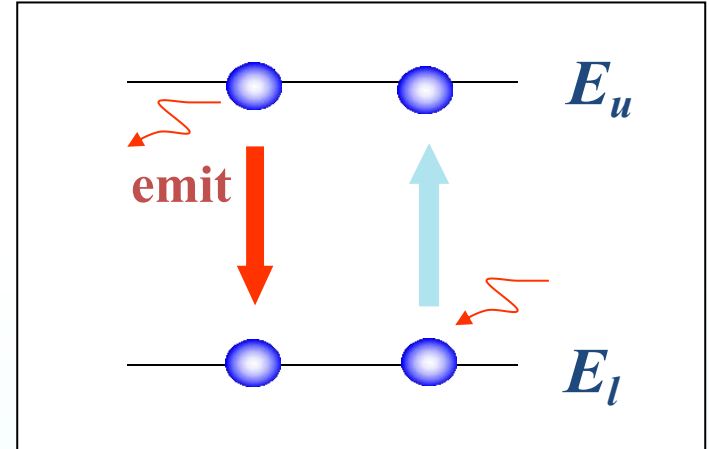


(b) Radiation assumption(跃迁假设)

$$hf = E_u - E_l$$

E_u : upper state

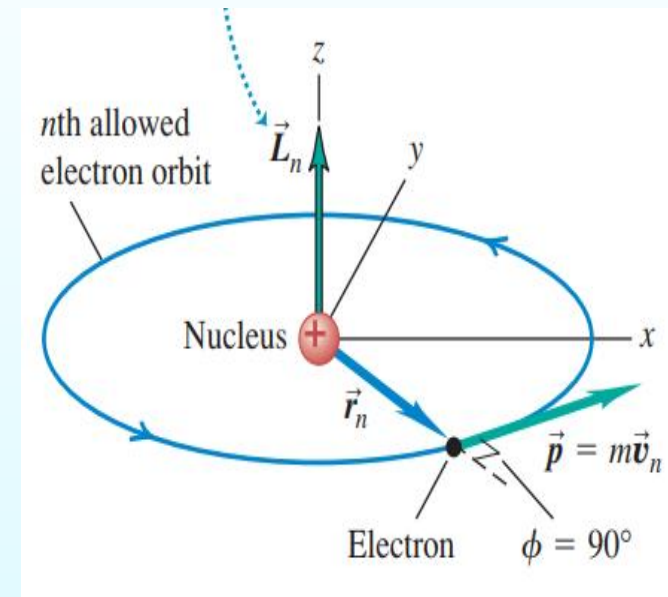
E_l : lower state



(c) Quantum condition (量子化条件)

Angular momentum of the stable orbits:

$$L = m v_n r_n = n \frac{h}{2\pi} = n\hbar \quad (h \text{ bar})$$



$n = 1, 2, 3, \dots$ n is called **quantum number**.

(2) Radius of Bohr's model

Electrical force $\frac{e^2}{4\pi\epsilon_0 r_n^2} = m \frac{v_n^2}{r_n}$

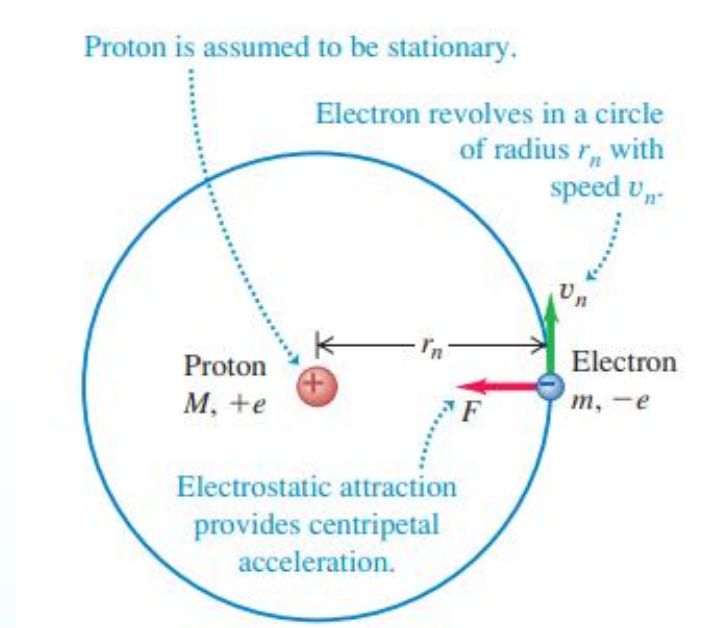
Quantum condition $m v_n r_n = n \frac{h}{2\pi}$

$$r_n = \frac{\epsilon_0 h^2}{\pi m e^2} n^2 = n^2 r_1 \quad (n = 1, 2, 3, \dots)$$

$n = 1, r_1 = \frac{\epsilon_0 h^2}{\pi m e^2} = 5.29 \times 10^{-11} \text{ m} = 0.53 \text{ \AA}$ **Bohr radius**

$r_n = n^2 r_1 \quad (n = 1, 2, 3, \dots)$ **excited states**

For hydrogenlike ions (Z) $r_n = \frac{\epsilon_0 h^2}{\pi m Z e^2} n^2 = \frac{n^2}{Z} r_1 = \frac{r_H}{Z}$ $r_n = \frac{r_H}{Z}$

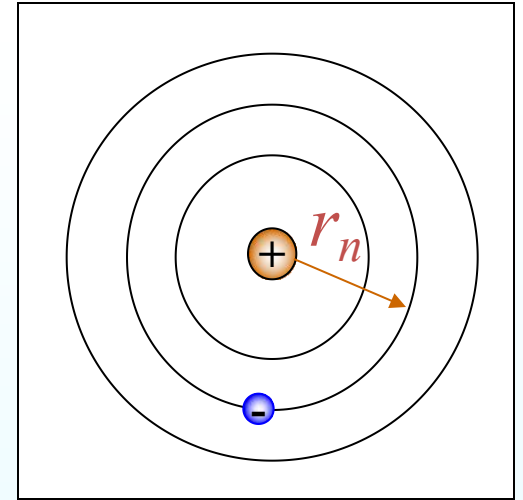


Correspondence principle

$$\Delta r = r_n - r_{n-1} = n^2 r_1 - (n-1)^2 r_1 = (2n-1)r_1$$

For large values of n

$$\Delta r \approx 2nr_1 = \frac{2r_n}{n} \quad \frac{\Delta r}{r_n} = \frac{2}{n}$$



So $\Delta r / r_n \rightarrow 0$, as $n \rightarrow \infty$

In accordance with the correspondence principle.

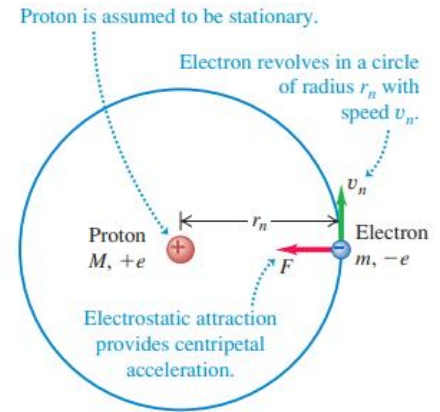
(3) Energy of Bohr's model

For n^{th} orbit
$$E_n = \frac{1}{2} m v_n^2 - \frac{e^2}{4 \pi \epsilon_0 r_n}$$

$$E_n = - \frac{e^2}{8 \pi \epsilon_0 r_n} = - \frac{m e^4}{8 \epsilon_0^2 h^2} \cdot \frac{1}{n^2} = \frac{E_1}{n^2}$$

$$z = 1, \quad E_1 = - \frac{m e^4}{8 \epsilon_0^2 h^2} = \boxed{-13.6 \text{ eV}}$$

$$\boxed{E_n = \frac{E_1}{n^2} \quad (n = 1, 2, 3, \dots)}$$



ground state

excited states

Ionization energy: the **minimum** energy required to remove an electron from **ground** state of an atom

For **hydrogenlike** ions (z):

$$E_n = \frac{1}{2} m v_n^2 - \frac{Z e^2}{4 \pi \epsilon_0 r_n} = Z^2 \frac{E_1}{n^2} = Z^2 E_H$$

$$\boxed{E_n = Z^2 E_H}$$

Correspondence principle

$$hf = \frac{1}{hc}(E_n - E_{n-1}) = |E_1| \left[\frac{1}{(n-1)^2} - \frac{1}{n^2} \right] = |E_1| \frac{2n-1}{n^2(n-1)^2}$$

For **large** values of n

$$hf \approx \left(\frac{me^4}{8\varepsilon_0^2 h^2} \right) \frac{2n}{n^4} = \frac{me^4}{4\varepsilon_0^2 h^2 n^3} \Rightarrow f = \frac{me^4}{4\varepsilon_0^2 h^3 n^3}$$

$$L = m\vec{v} \times \vec{r}_n = n \frac{h}{2\pi} \Rightarrow v = \frac{nh}{2\pi m r_n} \quad r_n = \frac{\varepsilon_0 h^2}{\pi m e^2} n^2$$

The frequency of the light emitted is equal to

$$\frac{v_n}{2\pi r_n} = \frac{me^4}{4\varepsilon_0^2 h^3 n^3} = f$$

From the classical theory for an electron revolving in a circular orbit,

$$T = \frac{2\pi r_n}{v}$$

$$f = \frac{1}{T} = \frac{v}{2\pi r_n}$$

Classically an accelerated charge radiates. For circular motion, the frequency of the radiation is the orbital frequency. This agrees with the Bohr predicting for large values of n , consistent with the correspondence principle.

4 . Derive Rydberg constant from Bohr's theory

$$\frac{1}{\lambda} = \frac{f}{c} = \frac{hf}{hc}$$

$$\frac{1}{\lambda} = R\left(\frac{1}{m^2} - \frac{1}{n^2}\right), m = 1, 2, 3 \dots$$

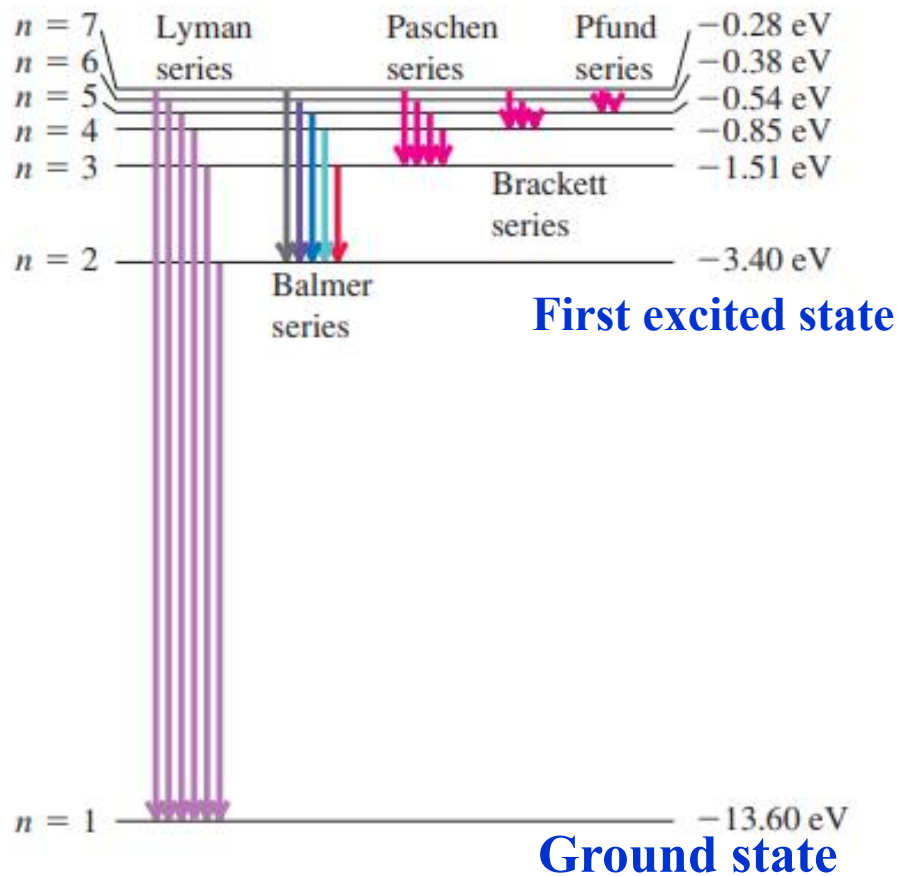
$$R = 1.0974 \times 10^7 \text{ m}^{-1} : \text{Rydberg constant}$$

$$= \frac{1}{hc} (E_n - E_m) = \frac{|E_1|}{hc} \left(\frac{1}{m^2} - \frac{1}{n^2} \right)$$

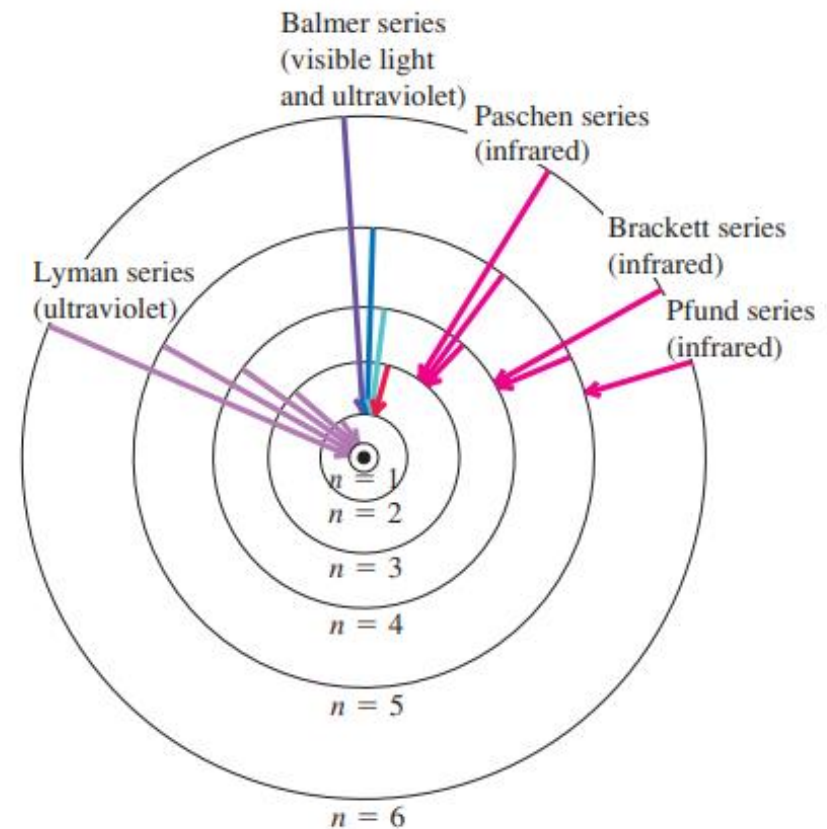
$$= \frac{me^4}{8\epsilon_0^2 h^3 c} \left(\frac{1}{m^2} - \frac{1}{n^2} \right), \quad n > m$$

$$\frac{me^4}{8\epsilon_0^2 h^3 c} = 1.097 \times 10^7 \text{ m}^{-1} \approx R$$

Energy-level diagram for hydrogen, showing some transitions corresponding to the various series



Permitted orbits of an electron in the Bohr model of a hydrogen atom (not to scale). Arrows indicate the transitions responsible for some of the lines of various series.



The advantages of Bohr's model

- (1) Bohr's model of the atom gave us a **first** (though rough) picture of what an atom is like.
- (2) He proposed the **stationary states**, which electrons in fixed orbits do not radiate light even though they are accelerating, and he assumed that angular **momentum** is **quantized**.
- (3) He said that an electron moved when it made a **transition** from one energy level to another and explained why there should be emission and absorption of light by atoms at discrete wavelengths, as well as for the stability of atoms.

Limitations of Bohr's models

(1) Bohr's theory worked well for hydrogen and for **one-electron** ions. But the theory could not predict line spectra for multielectron atoms, even for the next simplest atom, helium.

(2) It could not explain why some emission lines are **brighter** than others, nor why some lines are actually split into two or more closely spaced lines (fine structure).

(3) It was a strange **mixture** of classical (Electrical force , Newton's law) and quantum (Quantum condition) ideas .

Take the potential energy of a hydrogen atom to be zero for infinite separation of the electron and proton. Then the ground state energy is -13.6 eV . When the electron is in the first excited state the ionization energy is:



- A. 0
- ☒ B. 3.4 eV
- C. 6.8 eV
- D. 10.2 eV
- E. 13.6 eV

Example Wavelength of a Lyman line.

Determine the **wavelength** of the **first Lyman line**, the transition from $n=2$ to $n=1$. In what region of the electromagnetic spectrum does this line?

Solution:

$$hf = E_2 - E_1 = -3.4 - (-13.6)$$

$$= 10.2 \text{ eV} = 1.63 \times 10^{-18} \text{ J}$$

$$\lambda = \frac{c}{f} = \frac{hc}{E_2 - E_1}$$

$$= \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{1.63 \times 10^{-18}} = 1.22 \times 10^{-7} \text{ m}$$

Ultraviolet



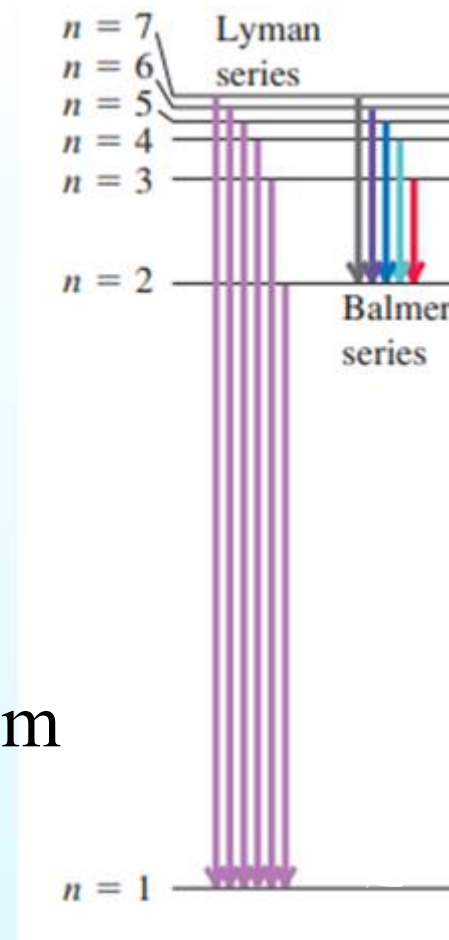
Example A hydrogen atom absorbs a photon, which causes an electron in a state corresponding to $n=2$ to jump to the state corresponding to $n=5$. Determine (a) the **energy** and (b) the **wavelength** of the absorbed photon.

Solution:

$$\begin{aligned}(a) \Delta E &= E_5 - E_2 = (-13.6) \times \left(\frac{1}{5^2} - \frac{1}{2^2} \right) \\ &= 2.86 \text{ eV} (4.57 \times 10^{-19} \text{ J})\end{aligned}$$

$$(b) \Delta E = hf = f \frac{c}{\lambda}$$

$$\lambda = \frac{hc}{\Delta E} = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{4.57 \times 10^{-19}} = 4.35 \times 10^{-7} \text{ m}$$



Example : Electrons accelerated by a potential difference of 12.6V pass through a gas of hydrogen atoms at room temperature, **how many** spectra lines will be appeared?

解:

$$\left. \begin{aligned} \Delta E &= E_n - E_1 \\ E_n &= -\frac{E_1}{n^2} \end{aligned} \right\} \begin{aligned} 12.6 &= -\frac{13.6}{n^2} - (-13.6) \\ n &= \sqrt{\frac{13.6}{13.6 - 12.6}} \approx 3.69 \quad \text{取 } n = 3 \end{aligned}$$

$$n = 3 \rightarrow n = 1 \quad h \frac{c}{\lambda_1} = \frac{E_1}{3^2} - \frac{E_1}{1^2} \quad \lambda_1 = 1.025 \times 10^{-7} \text{ m}$$

$$n = 3 \rightarrow n = 2 \quad h \frac{c}{\lambda_2} = \frac{E_1}{3^2} - \frac{E_1}{2^2} \quad \lambda_2 = 6.579 \times 10^{-7} \text{ m}$$

$$n = 2 \rightarrow n = 1 \quad h \frac{c}{\lambda_3} = \frac{E_1}{2^2} - \frac{E_1}{1^2} \quad \lambda_3 = 1.216 \times 10^{-7} \text{ m}$$

22. 5 Matter waves

1. The nature of matter

If light has the **wave-particle duality**, those things in nature thought to be particles might also have **wave properties**.

Particle

Energy E

$$E = mc^2$$

$$E = hf$$

Momentum p

$$P = mv$$

$$p = \frac{h}{\lambda}$$

Wave

frequency f

$$f = \frac{c}{\lambda}$$

wavelength λ

$$\lambda = \frac{c}{f}$$



de Broglie



$$f = \frac{E}{h} = \frac{mc^2}{h} = \frac{m_0 c^2}{h \sqrt{1 - v^2 / c^2}}$$

de Broglie wavelength

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{m_0 v} \sqrt{1 - v^2 / c^2} \quad \text{if } v \ll c, \quad \lambda = \frac{h}{m_0 v}$$

The relation holds for **all particles**, like photon, electron and so on. The wave of particles with nonzero mass is called **matter wave**.

Principle of complementary: We must use either the **particle** or **wave** property to interpret any given experiment. But we must be aware of **both** the particle and wave aspects in order to have an understanding of matter.

Consider the following three particles:

- 1. a free electron with kinetic energy K_0**
- 2. a free proton with kinetic energy K_0**
- 3. a free proton with kinetic energy $2K_0$**



Rank them according to the **wavelengths of their matter waves, least to greatest.**

A. 1, 2, 3

B. 3, 2, 1

C. 2, 3, 1

D. 1, 3, 2

E. 1, then 2 and 3 tied

Example Calculate the **de Broglie wavelength** of a 0.20-kg ball moving with a speed of 15m/s.

Solution:

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{0.20 \times 15} = 2.2 \times 10^{-34} m$$

For **macro** objects, their wavelengths are small and the wave properties go undetected.

德布罗意波波长很小，一般不显示波动性.

Example Electrons with charge e , rest mass m_0 is accelerated by voltage U .

Determine λ =?

Solution: $v \ll c, \quad eU = \frac{1}{2}m_0v^2 \Rightarrow v = \sqrt{2eU/m_0}$

$$\lambda = \frac{h}{p} = \frac{h}{m_0v} = \frac{h}{\sqrt{2em_0U}} \approx \frac{12.25}{\sqrt{U}} \times 10^{-10} m = \frac{1.225}{\sqrt{U}} nm$$

$$U = 150V \quad \lambda \approx 0.1nm$$

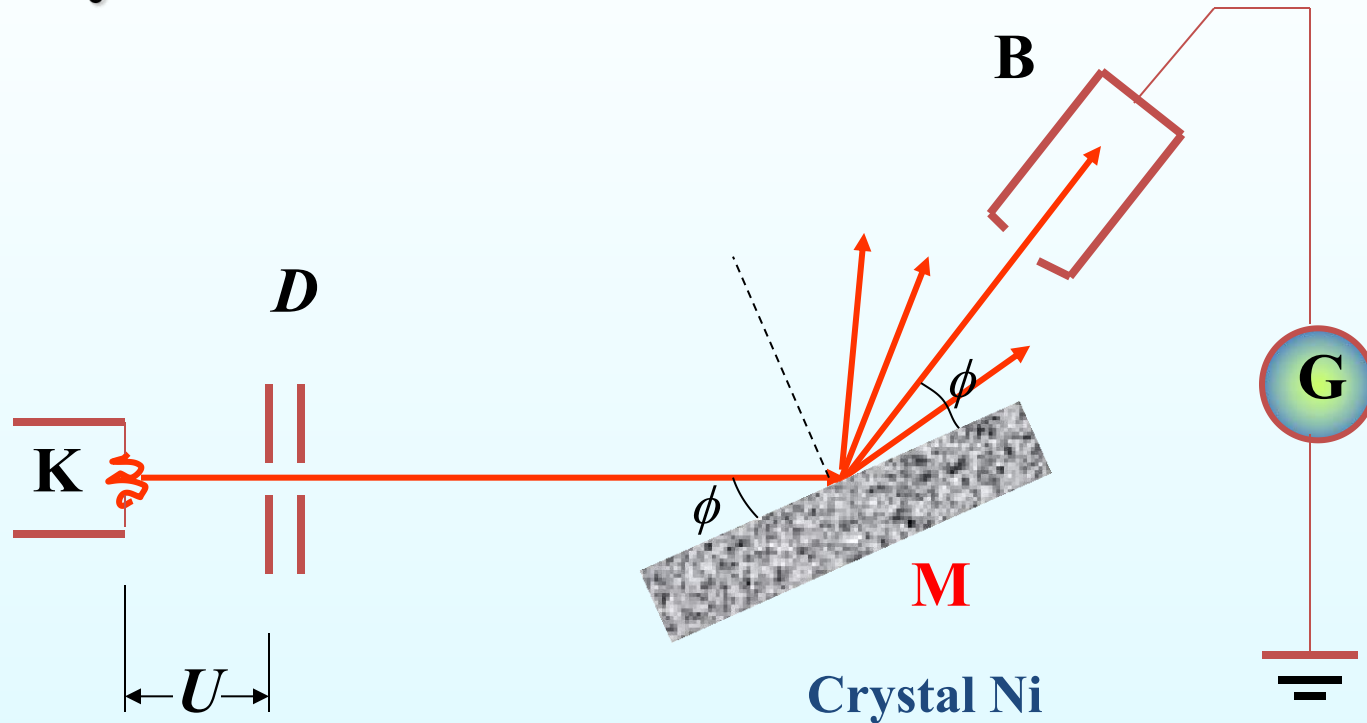
$$U = 10^4V \quad \lambda \approx 0.0122nm$$

The size of electrons is in order $\text{\AA}$ and the wave property can be measured.

2. Experiments of electron diffraction

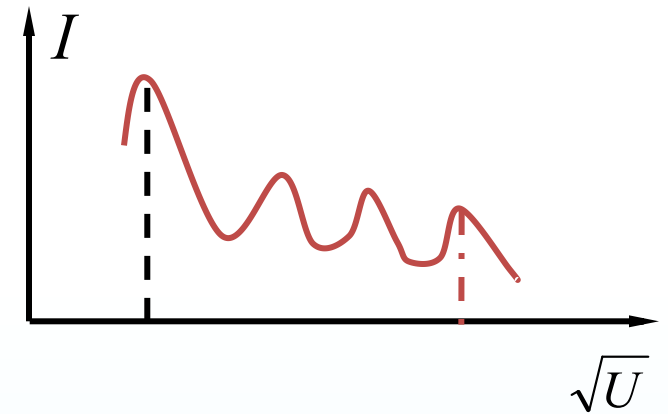
(1) Experiment of electron diffraction by C,J Davisson and L.H. Germer in Bell Lab 1927

Keep ϕ unchanged, increasing U , detecting current created by the electrons scattered from the metal Ni.



Bragg equation :

$$2d \sin \phi = m\lambda \quad (m = 1, 2, 3 \dots)$$



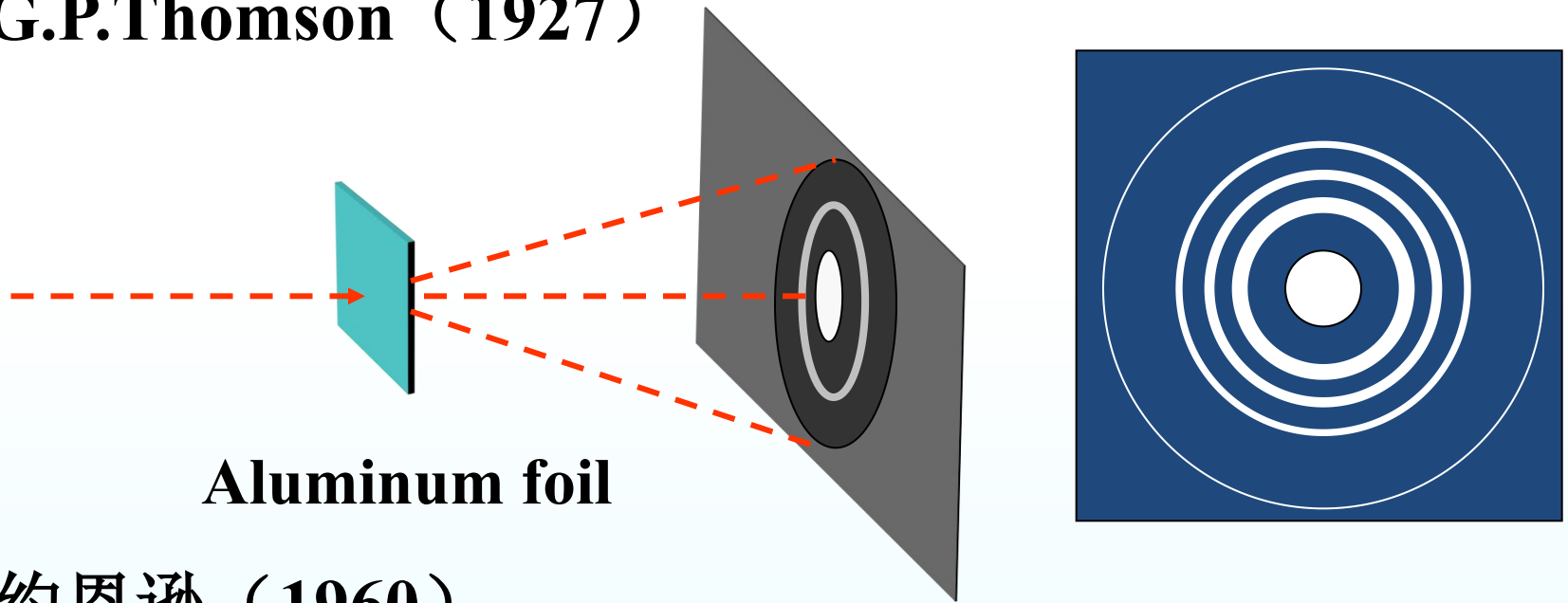
In the experiment:

$$d = 9.1 \times 10^{-11} \text{ m} \quad \phi = 65^\circ$$

$$\lambda = \frac{1.225}{\sqrt{U}} \text{ nm} \quad 2d \sin \phi = m \frac{1.225}{\sqrt{U}}$$

Electrons have **wave property.**

(2) G.P.Thomson (1927)



(3) 约恩逊 (1960)

电子的单缝、双缝、三缝和四缝衍射实验图象



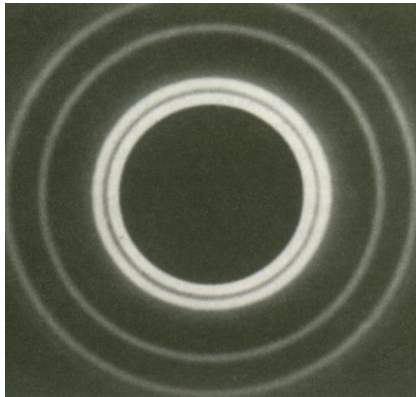
单缝衍射

双缝衍射

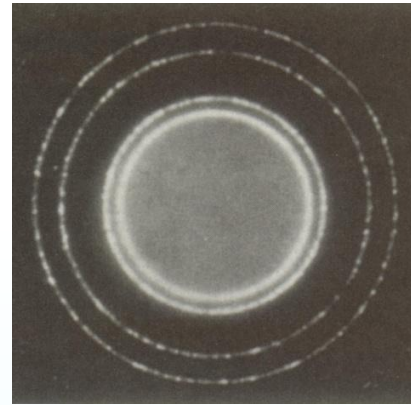
三缝衍射

四缝衍射

X- rays



***Diffraction
pattern***

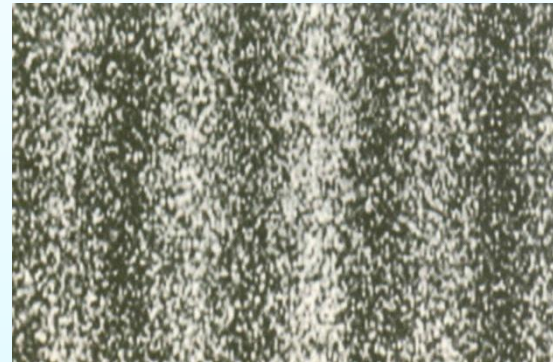


**Electron
beam**

$$\lambda = \frac{h}{p} = \frac{h}{m_0 v} \sqrt{1 - v^2 / c^2}$$



**Yong's double-slit
experiment**

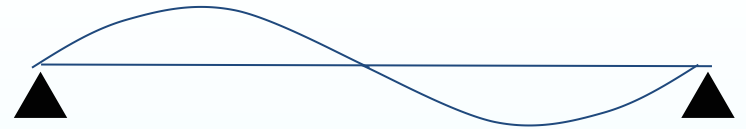


**Double-slit experiment
for electron beam**

3. de Broglie's Hypothesis applied to atoms

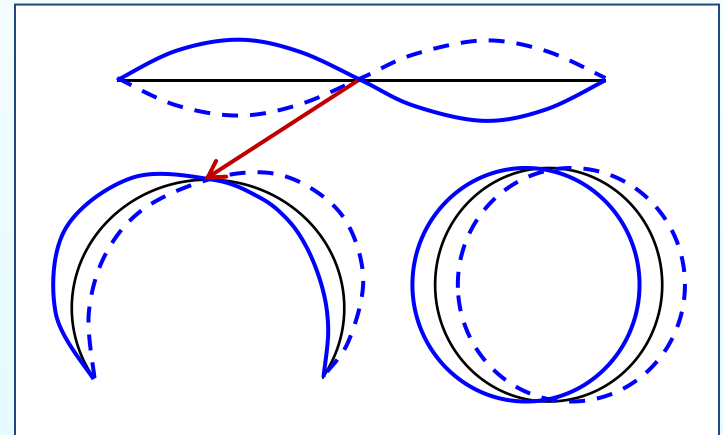
Standing wave (with two fixed ends)

$$L = n \frac{\lambda_n}{2} \quad (n = 1, 2, 3 \dots)$$



Curl the line and form a **circular orbit**

$$2 \pi r_n = n \lambda_n \quad (n = 1, 2, 3 \dots)$$



The orbit should contain an **integer** number of wavelengths.

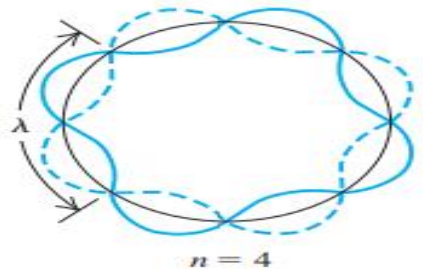
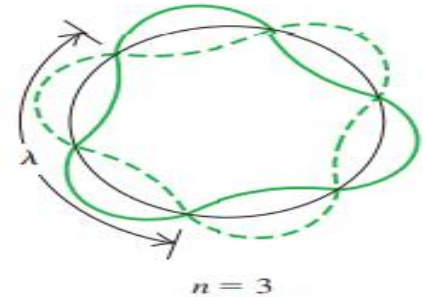
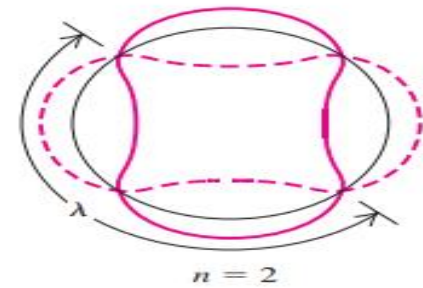
$$2 \pi r_n = n \lambda_n \quad (n = 1, 2, 3 \dots)$$

$$\lambda = \frac{h}{p} = \frac{h}{m v}$$

$$2 \pi r_n m v = n h$$

Quantum condition

$$L = m v r_n = n \frac{h}{2 \pi} = n \hbar$$



The quantized orbits and energy states in the Bohr's model are due to the wave nature of the electron. Only “resonant” **standing waves** can persist and the **wave-particle duality** is at the root of atomic structure.